

# BFS Traversal

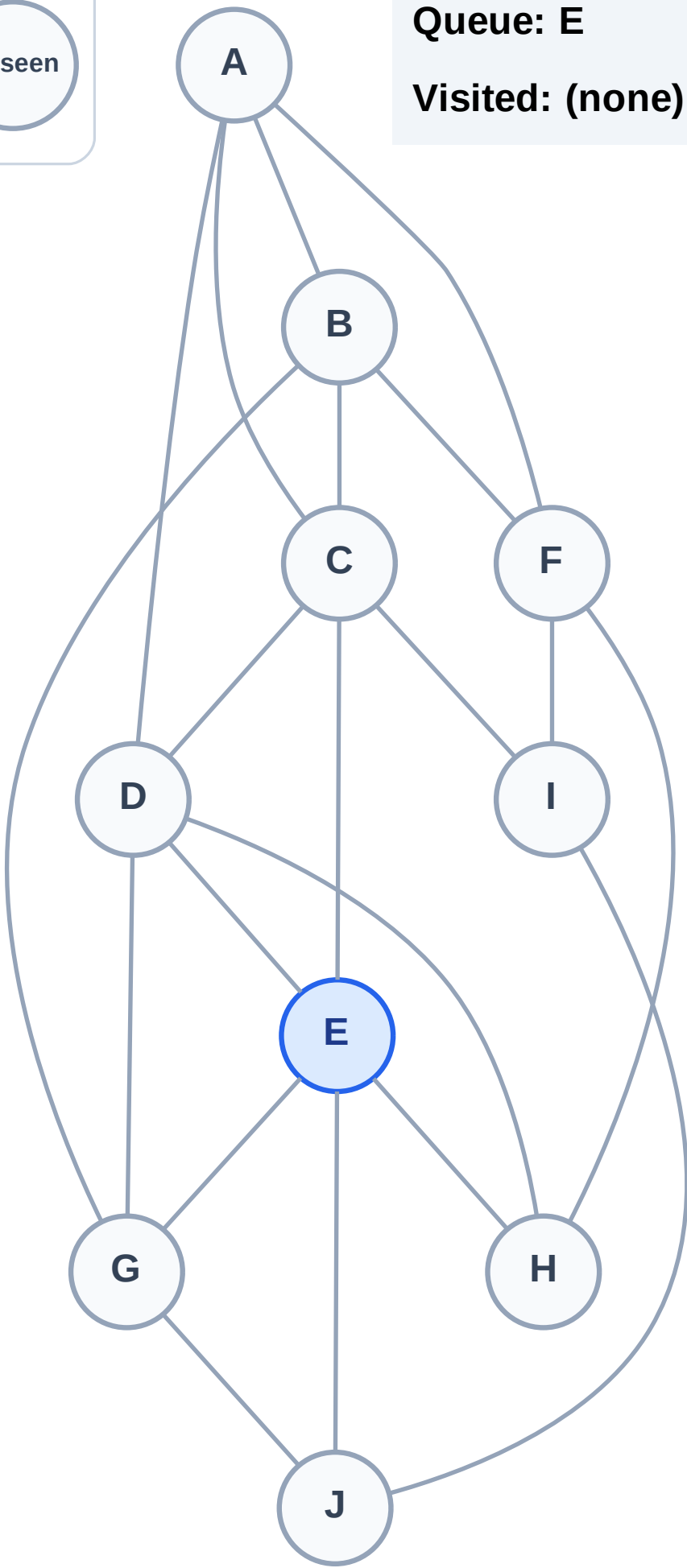
Step 1: Start at E

## Legend



Queue: E

Visited: (none)



# BFS Traversal

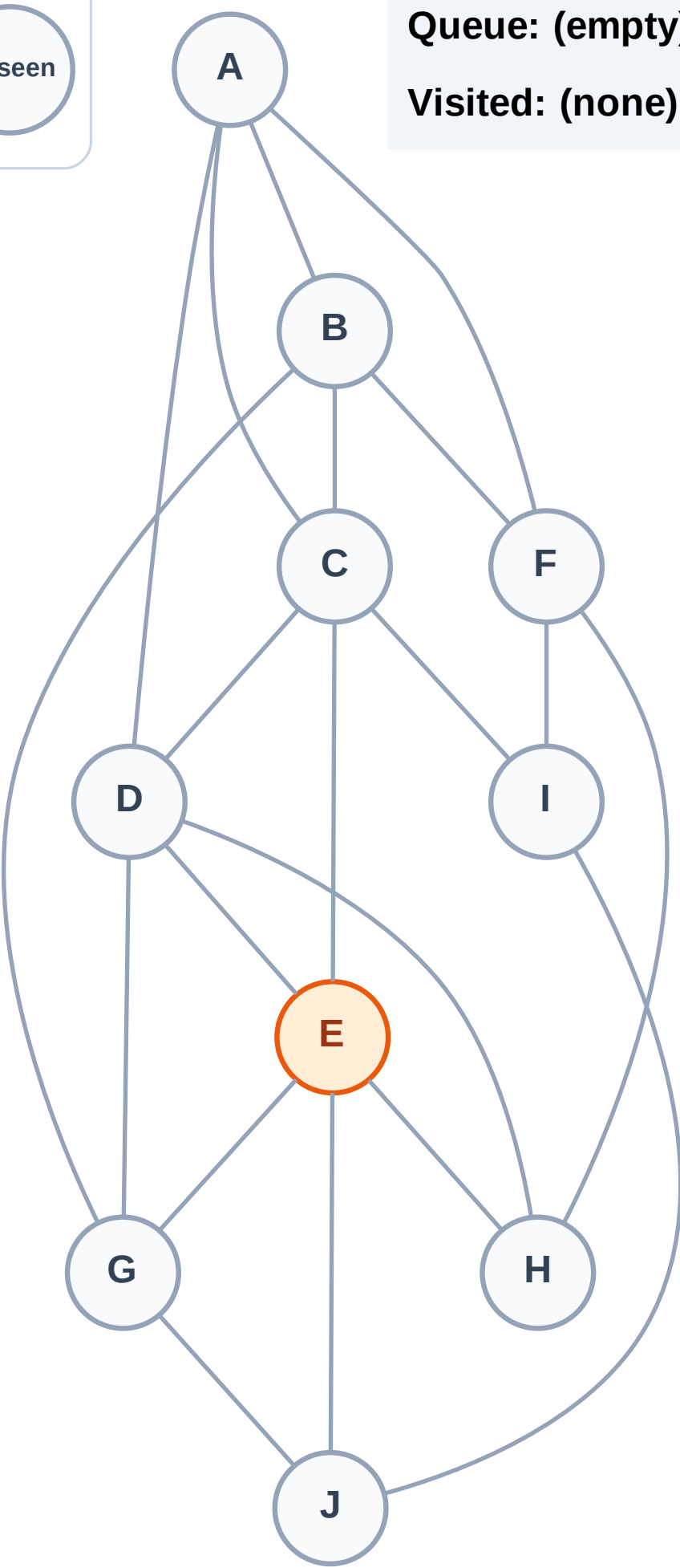
Step 2: Process node E

Legend



Queue: (empty)

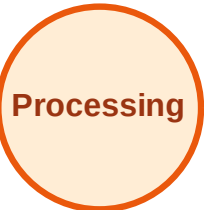
Visited: (none)



# BFS Traversal

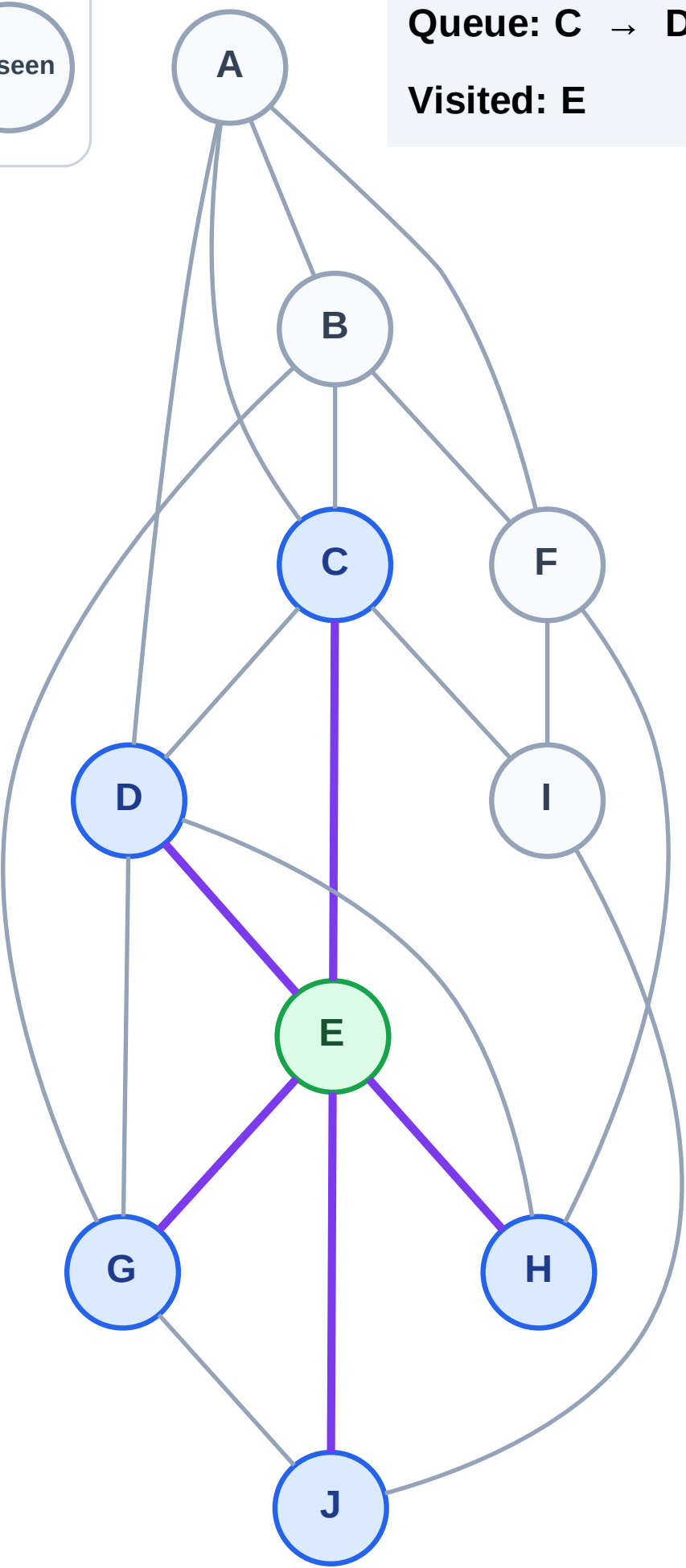
Step 3: Discover C, D, G, H, J from E

## Legend



Queue: C → D → G → H → J

Visited: E



# BFS Traversal

Step 4: Process node C

Legend

Complete

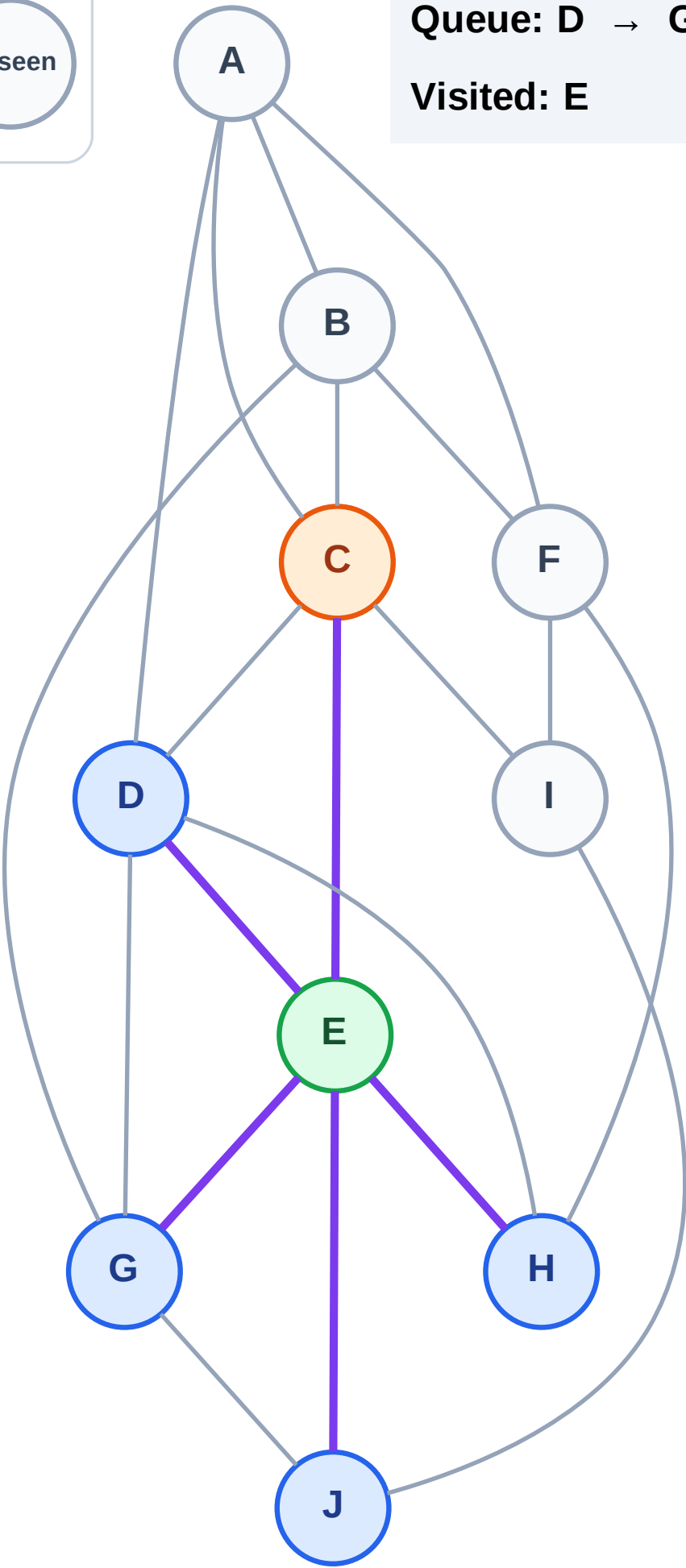
Processing

Discovered

Unseen

Queue: D → G → H → J

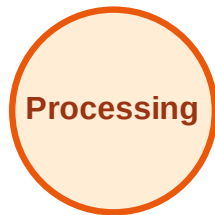
Visited: E



# BFS Traversal

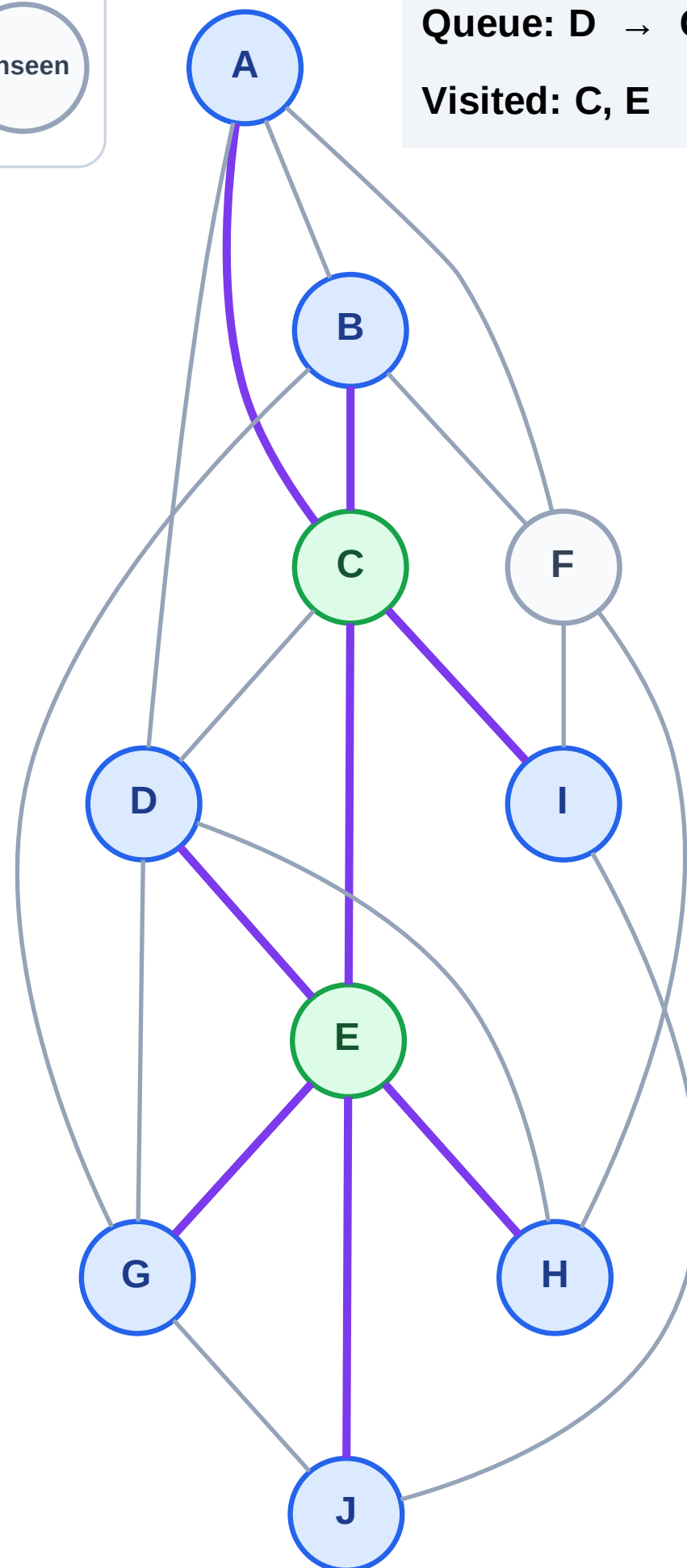
Step 5: Discover A, B, I from C

## Legend



Queue: D → G → H → J → A → B → I

Visited: C, E



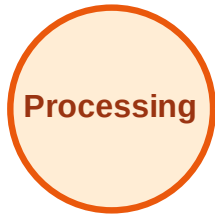
# BFS Traversal

Step 6: Process node D

## Legend



Complete



Processing



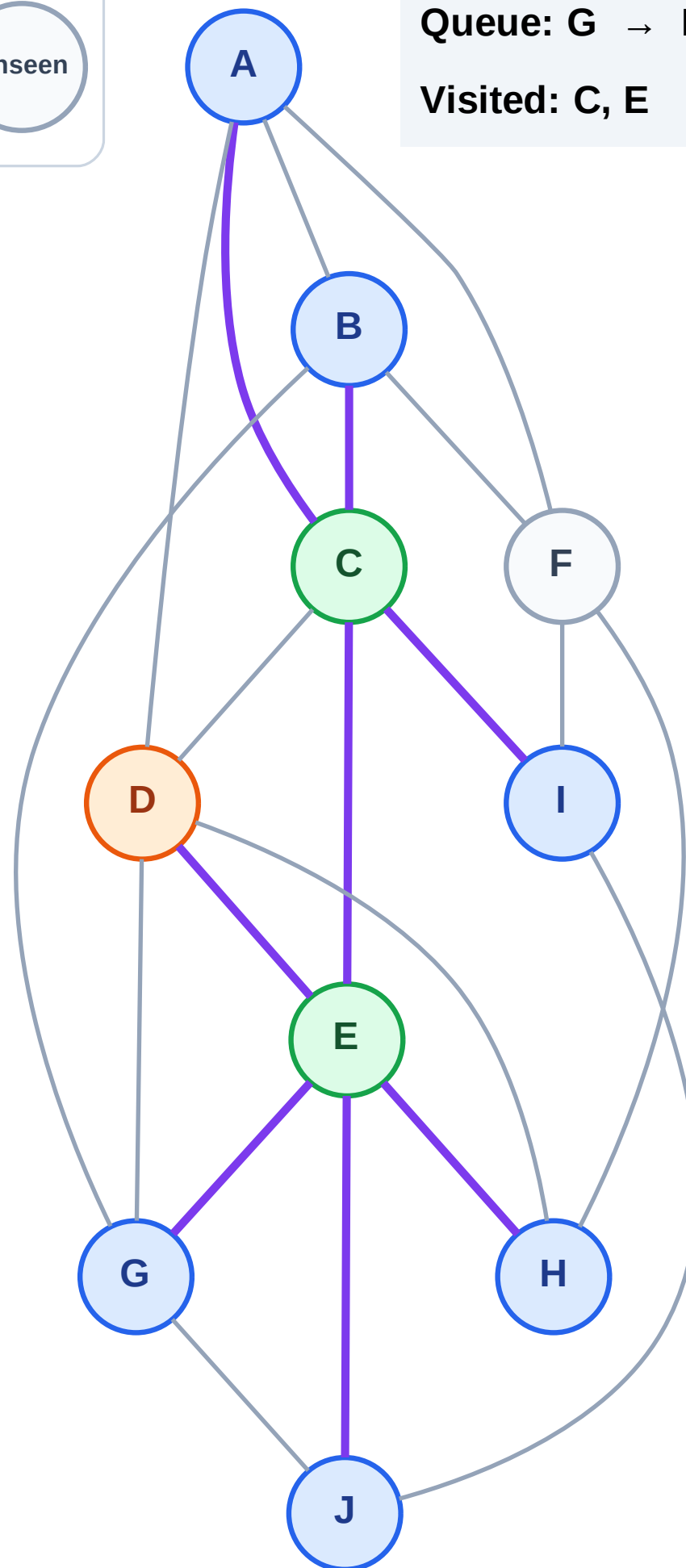
Discovered



Unseen

Queue: G → H → J → A → B → I

Visited: C, E





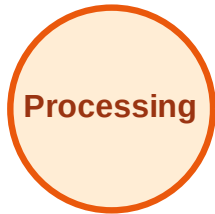
# BFS Traversal

Step 7: No new neighbors from D

## Legend



Complete



Processing



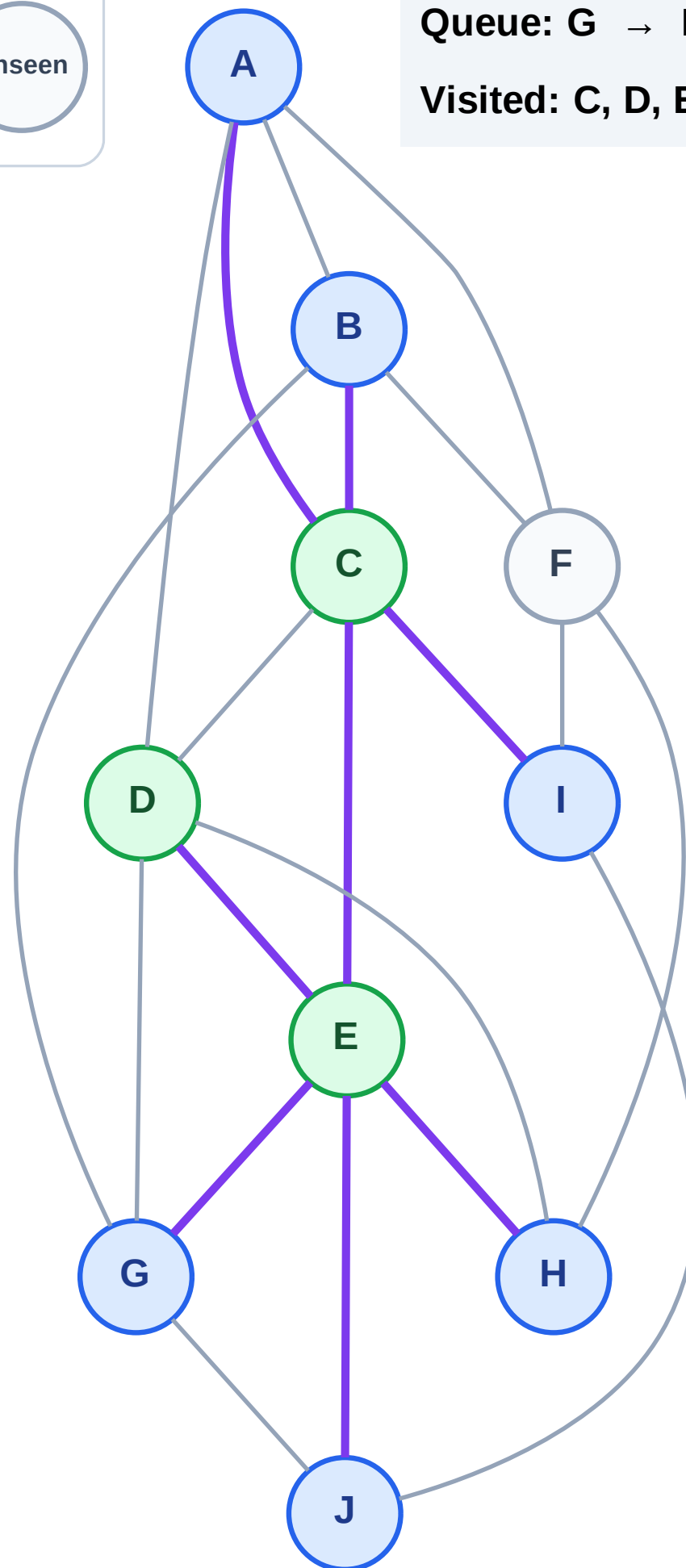
Discovered



Unseen

Queue: G → H → J → A → B → I

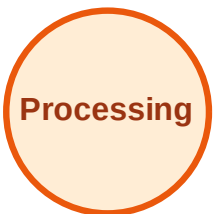
Visited: C, D, E



# BFS Traversal

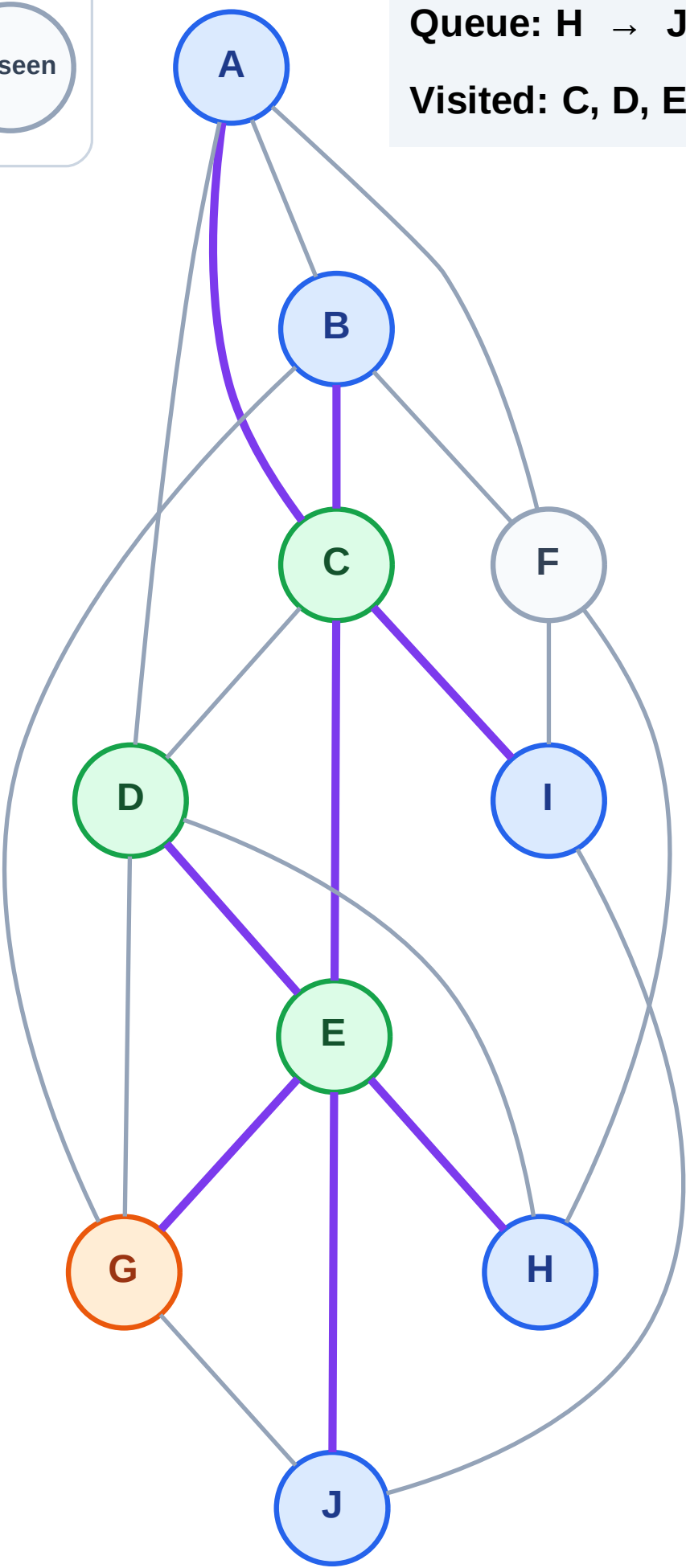
Step 8: Process node G

## Legend



Queue: H → J → A → B → I

Visited: C, D, E

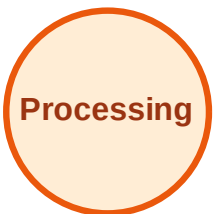




# BFS Traversal

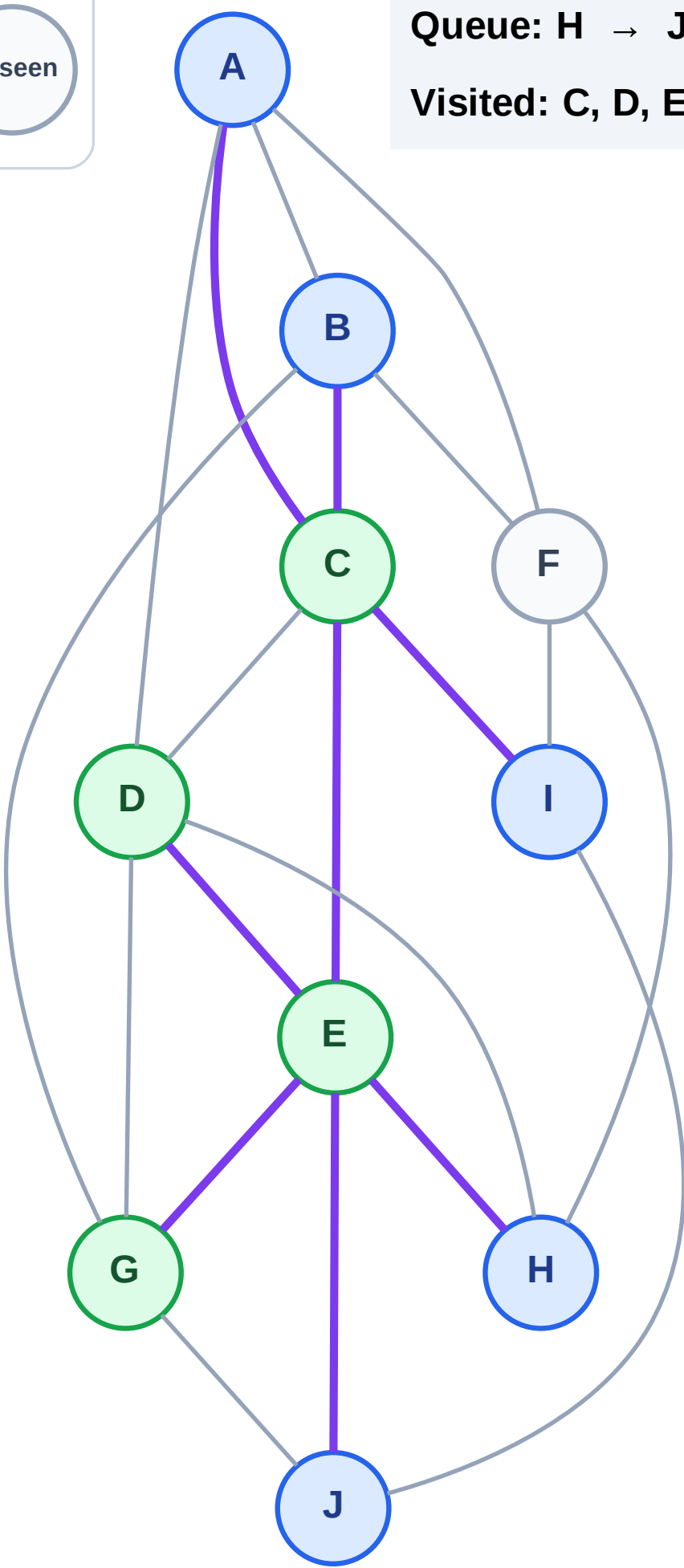
Step 9: No new neighbors from G

## Legend



Queue: H → J → A → B → I

Visited: C, D, E, G



# BFS Traversal

Step 10: Process node H

Legend

Complete

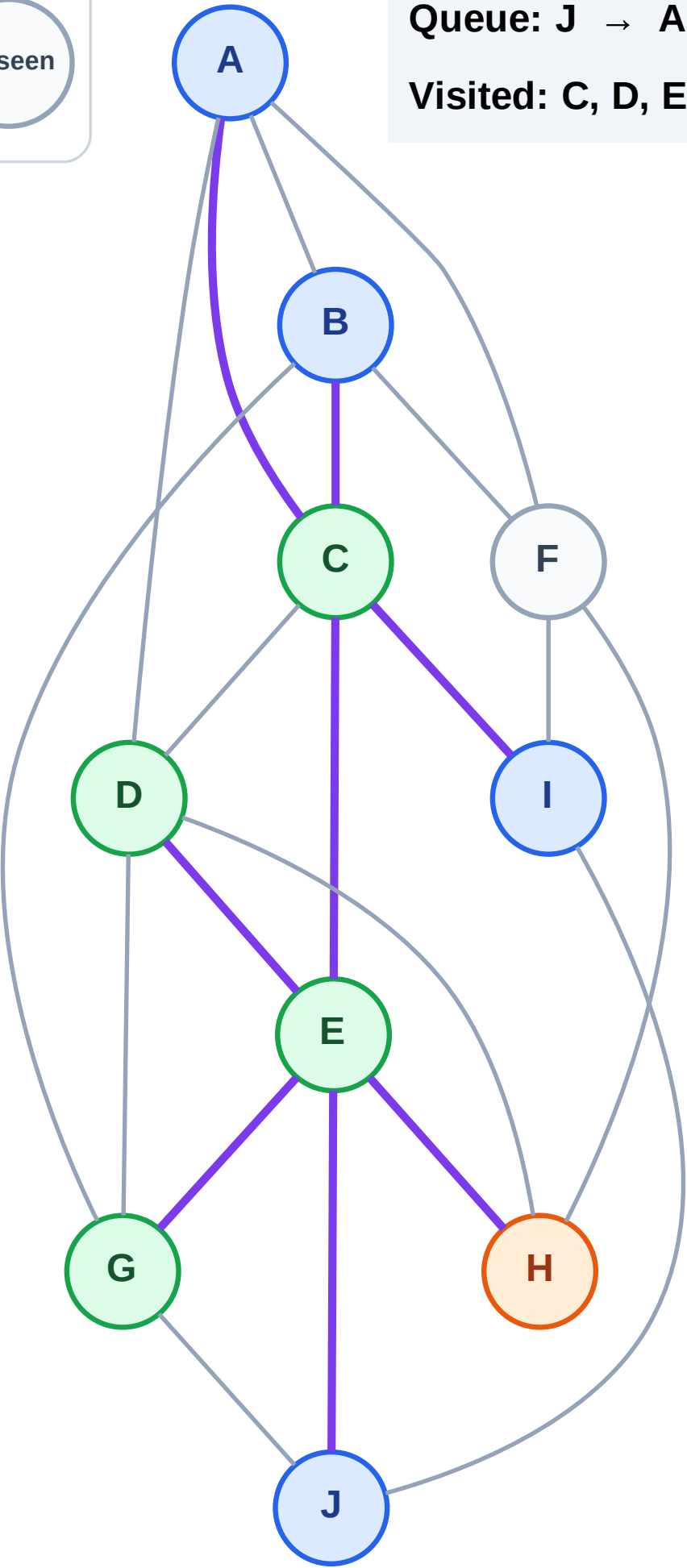
Processing

Discovered

Unseen

Queue: J → A → B → I

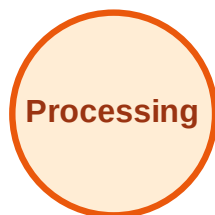
Visited: C, D, E, G



# BFS Traversal

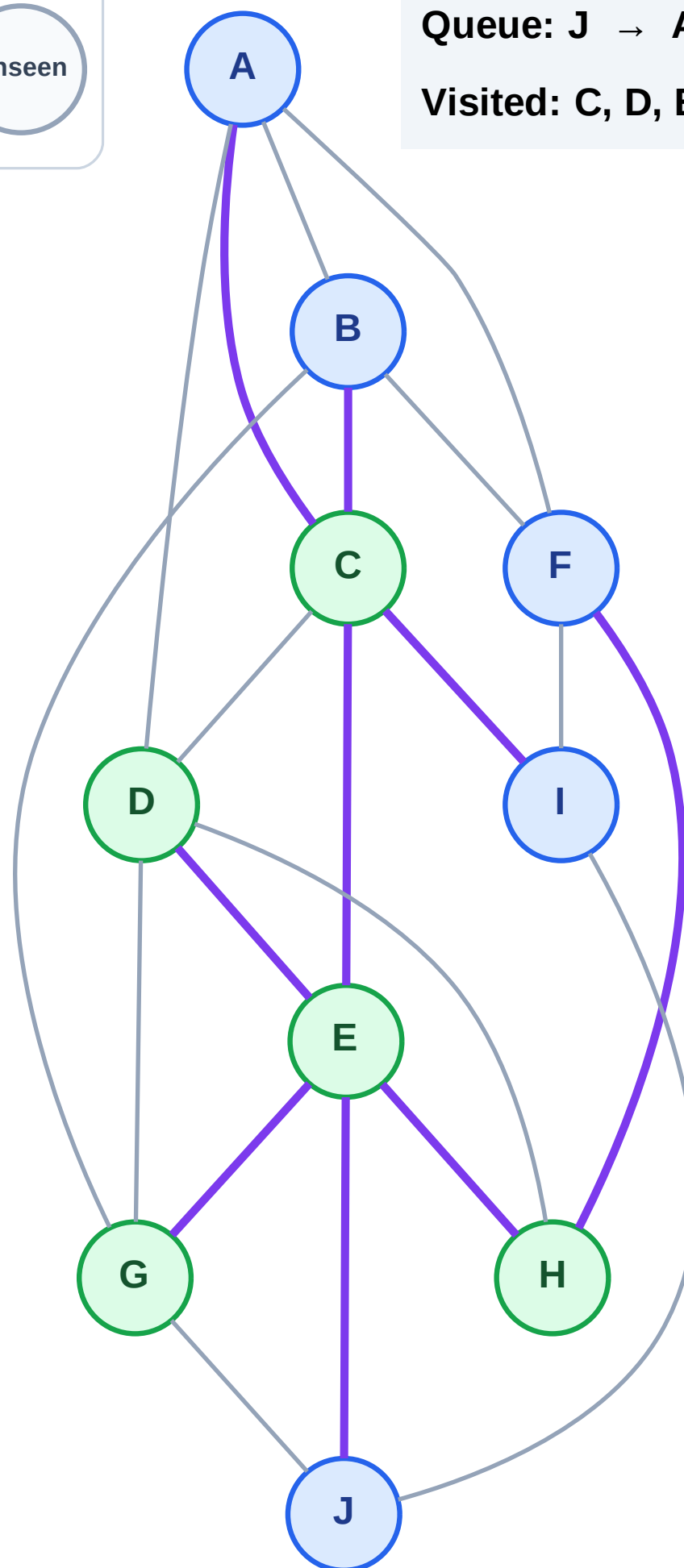
Step 11: Discover F from H

## Legend



Queue: J → A → B → I → F

Visited: C, D, E, G, H



# BFS Traversal

Step 12: Process node J

Legend

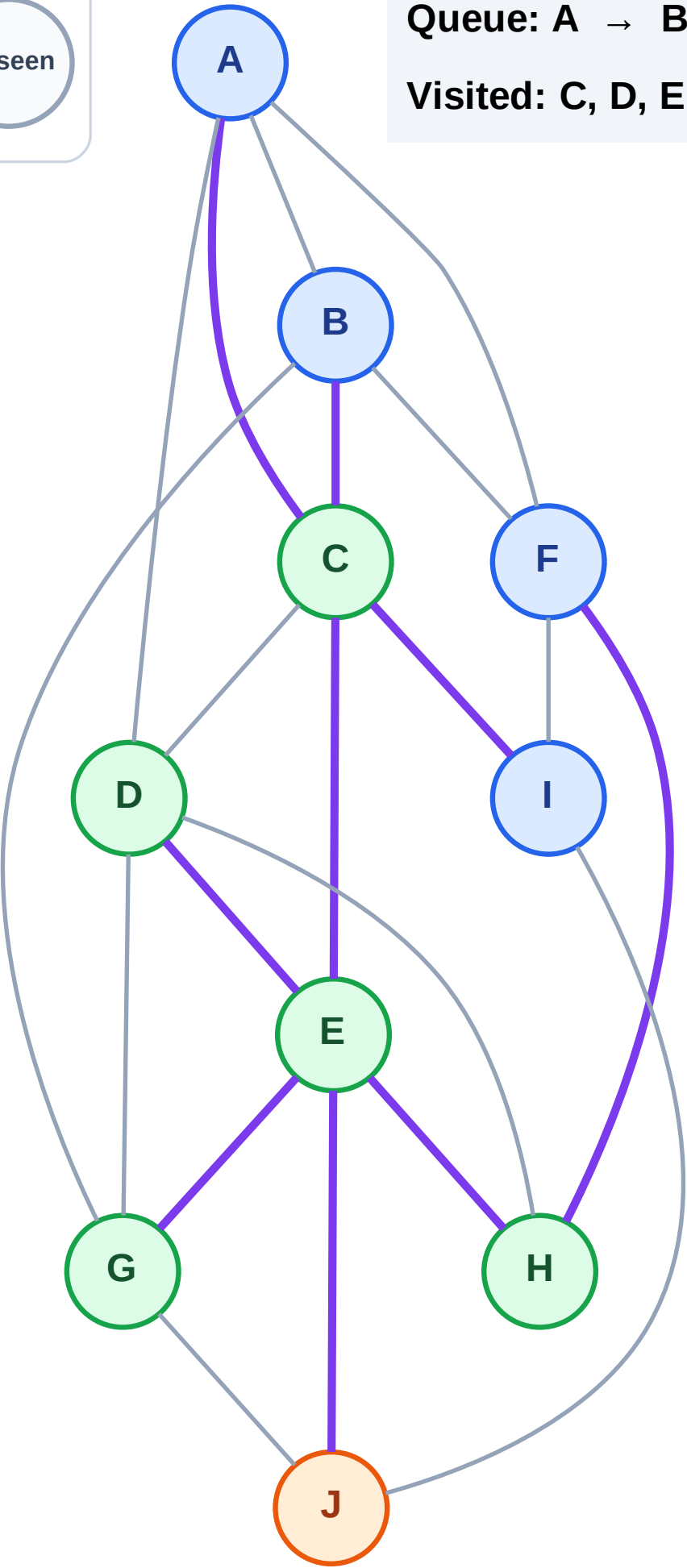
Complete

Processing

Discovered

Unseen

Queue: A → B → I → F  
Visited: C, D, E, G, H



# BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

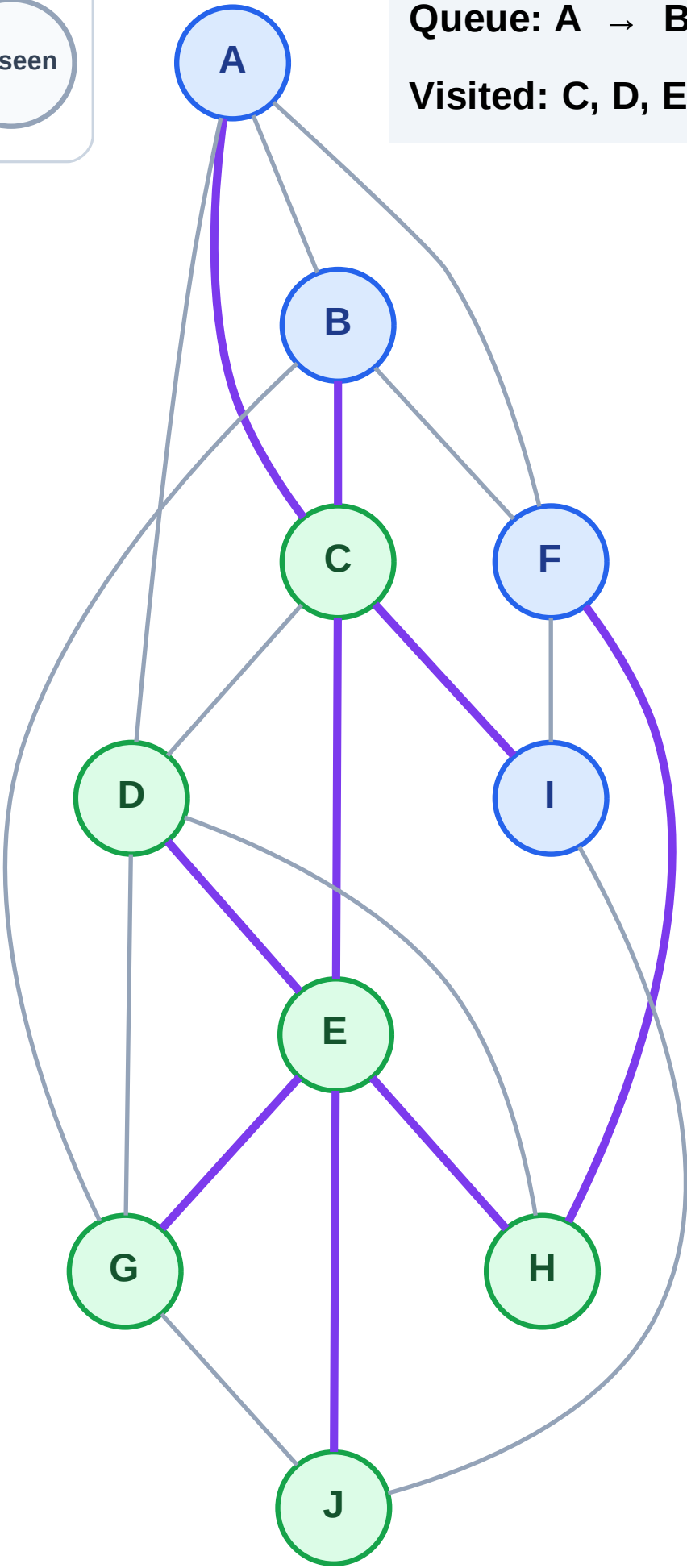
Processing

Discovered

Unseen

Queue: A → B → I → F

Visited: C, D, E, G, H, J



# BFS Traversal

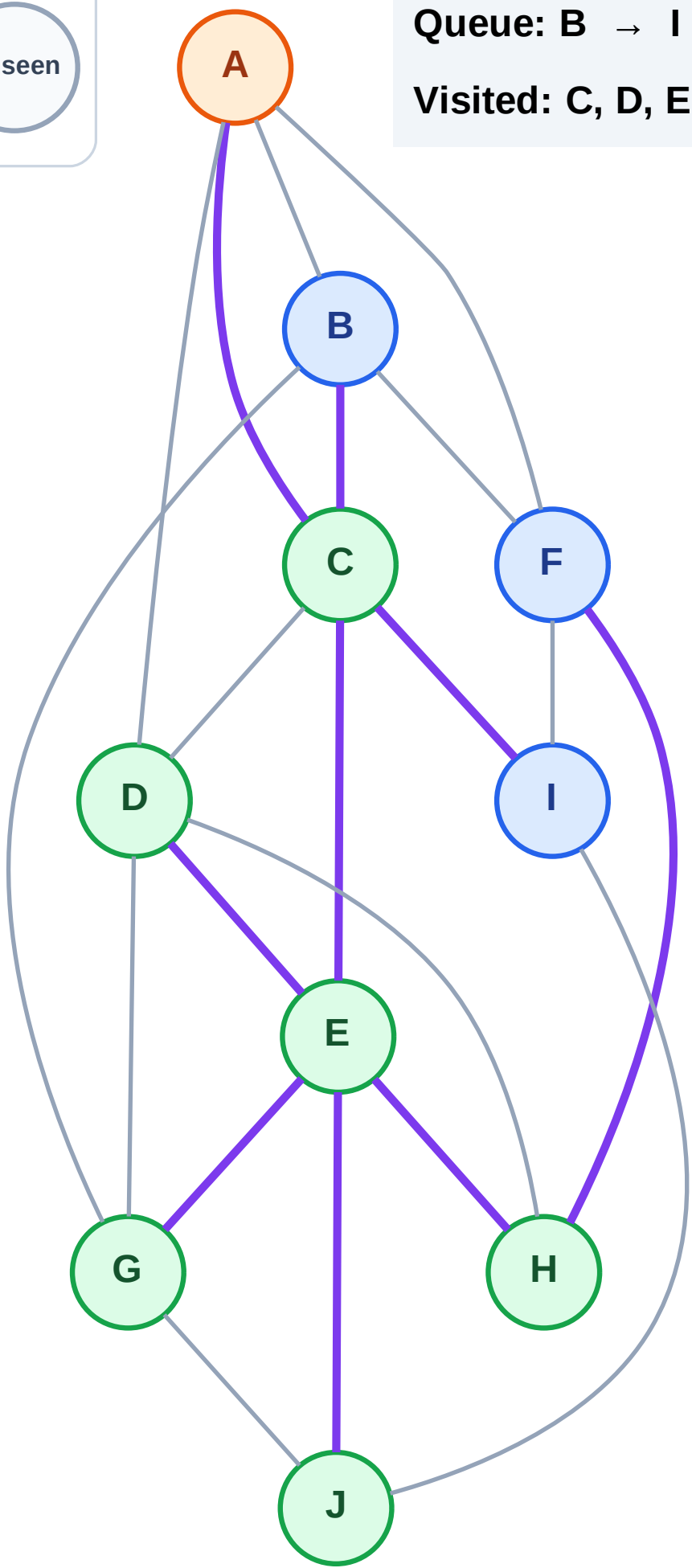
Step 14: Process node A

## Legend



Queue: B → I → F

Visited: C, D, E, G, H, J

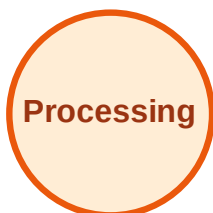




# BFS Traversal

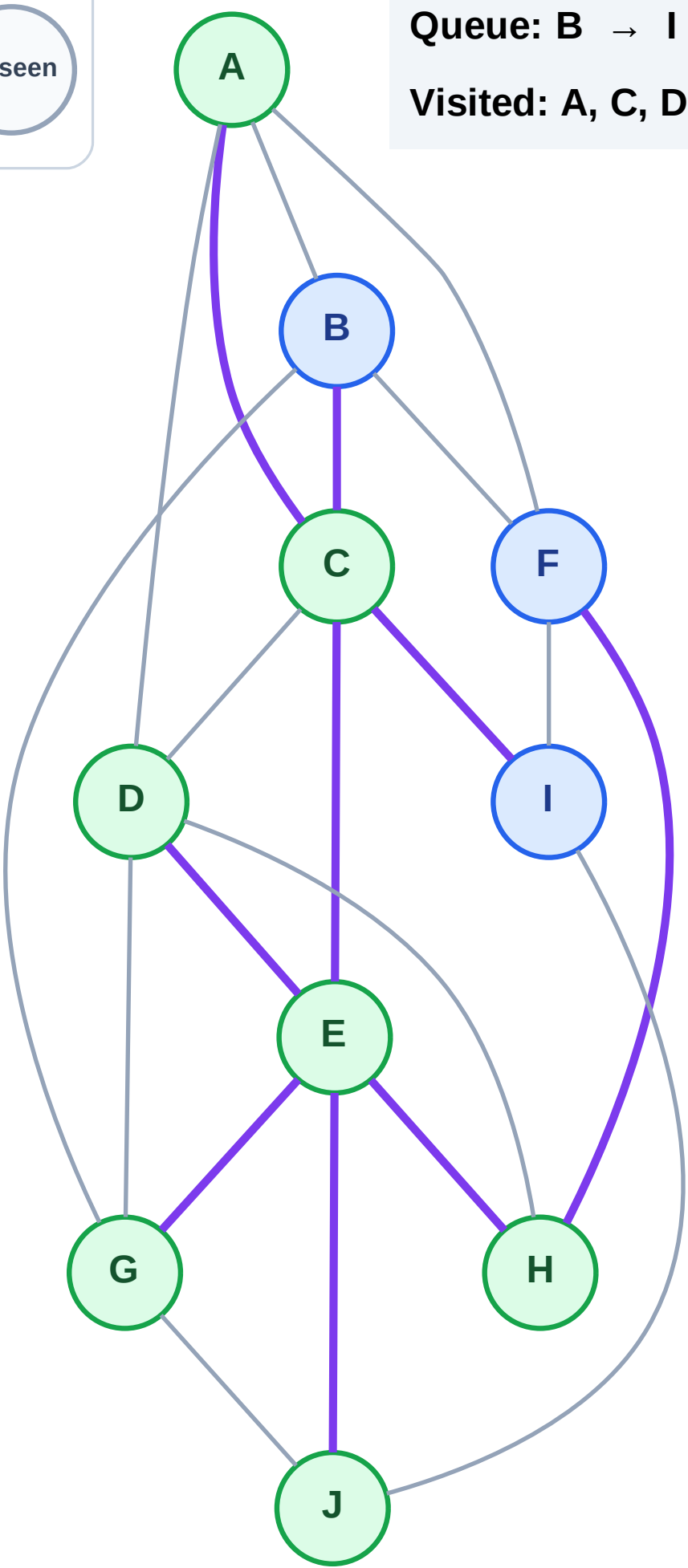
Step 15: No new neighbors from A

## Legend



Queue: B → I → F

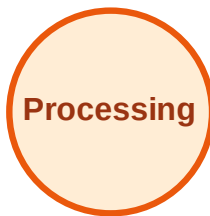
Visited: A, C, D, E, G, H, J



# BFS Traversal

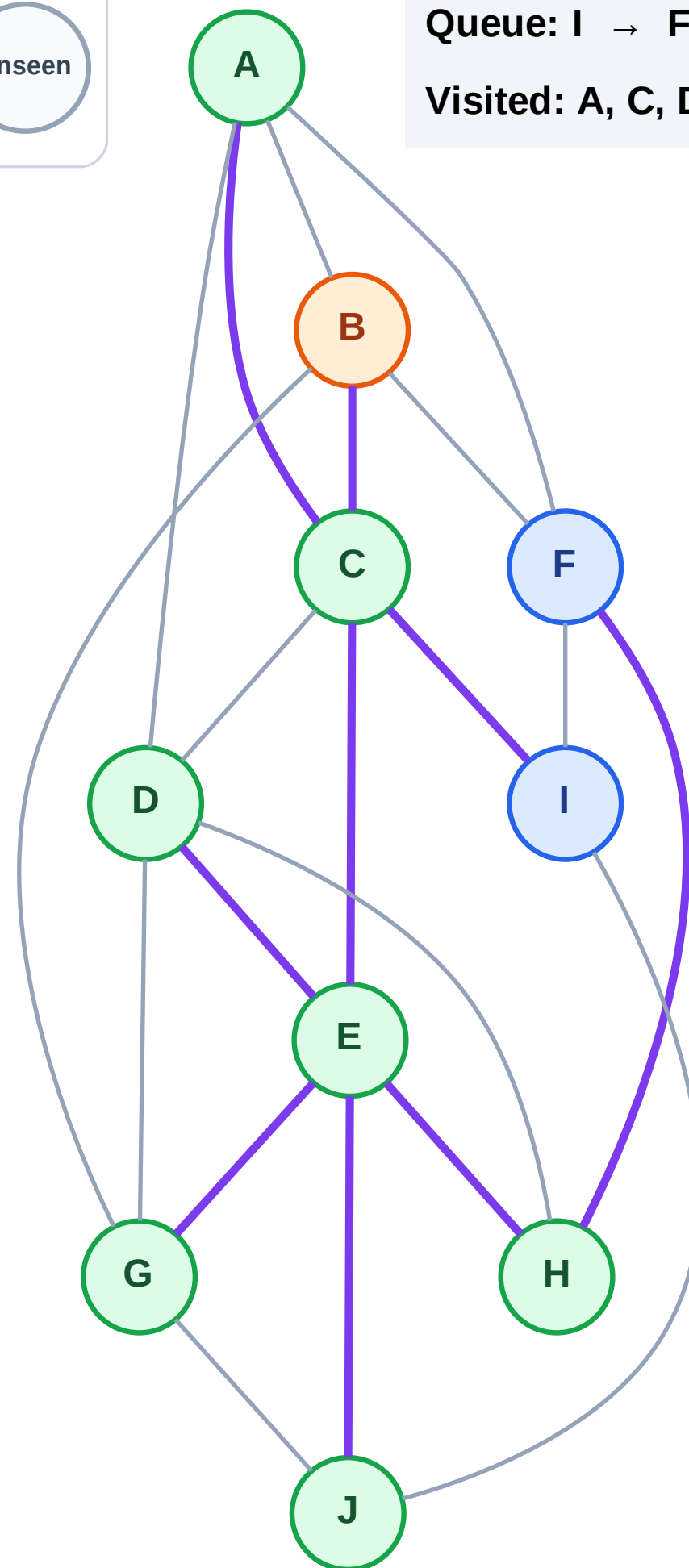
Step 16: Process node B

## Legend



Queue: I → F

Visited: A, C, D, E, G, H, J

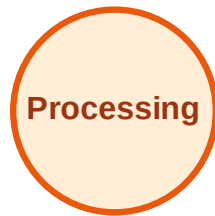




# BFS Traversal

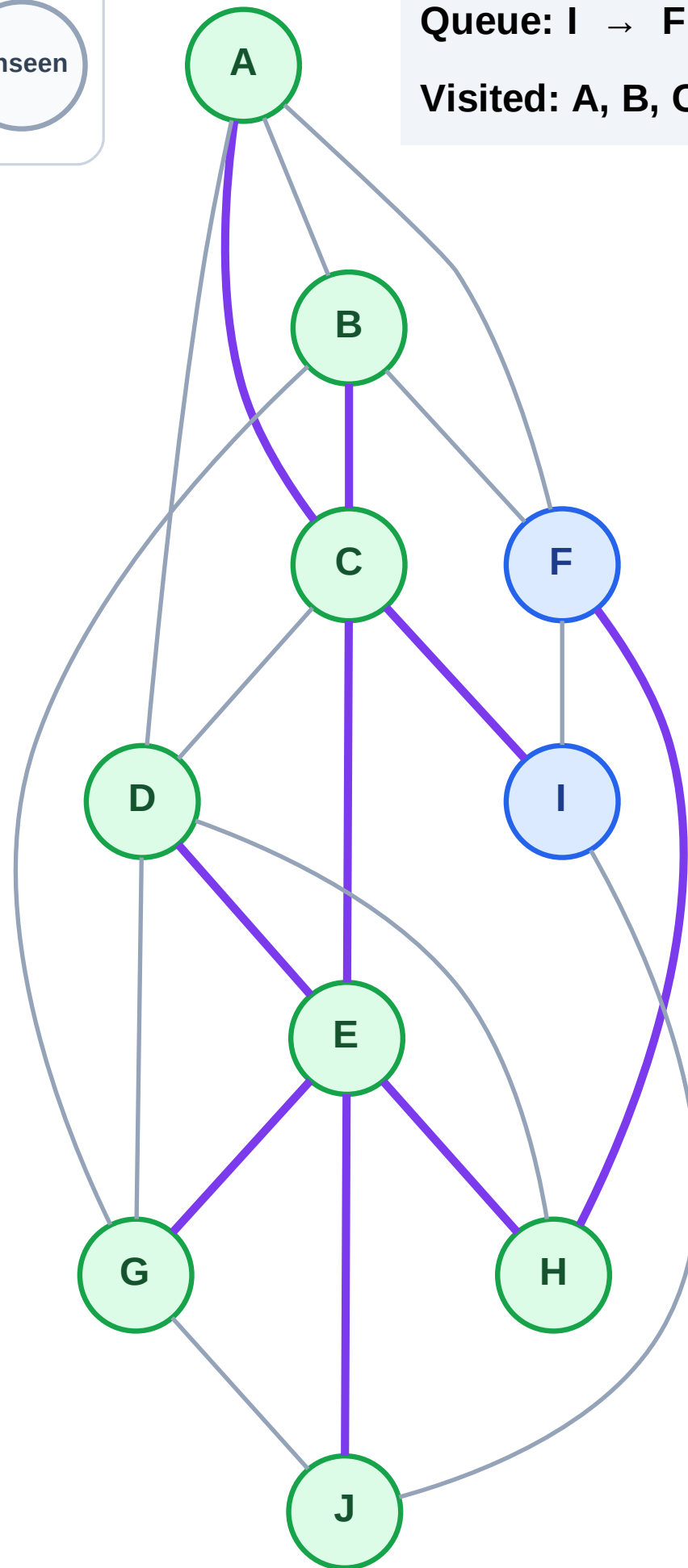
Step 17: No new neighbors from B

## Legend



Queue: I → F

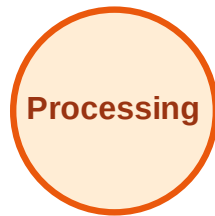
Visited: A, B, C, D, E, G, H, J



# BFS Traversal

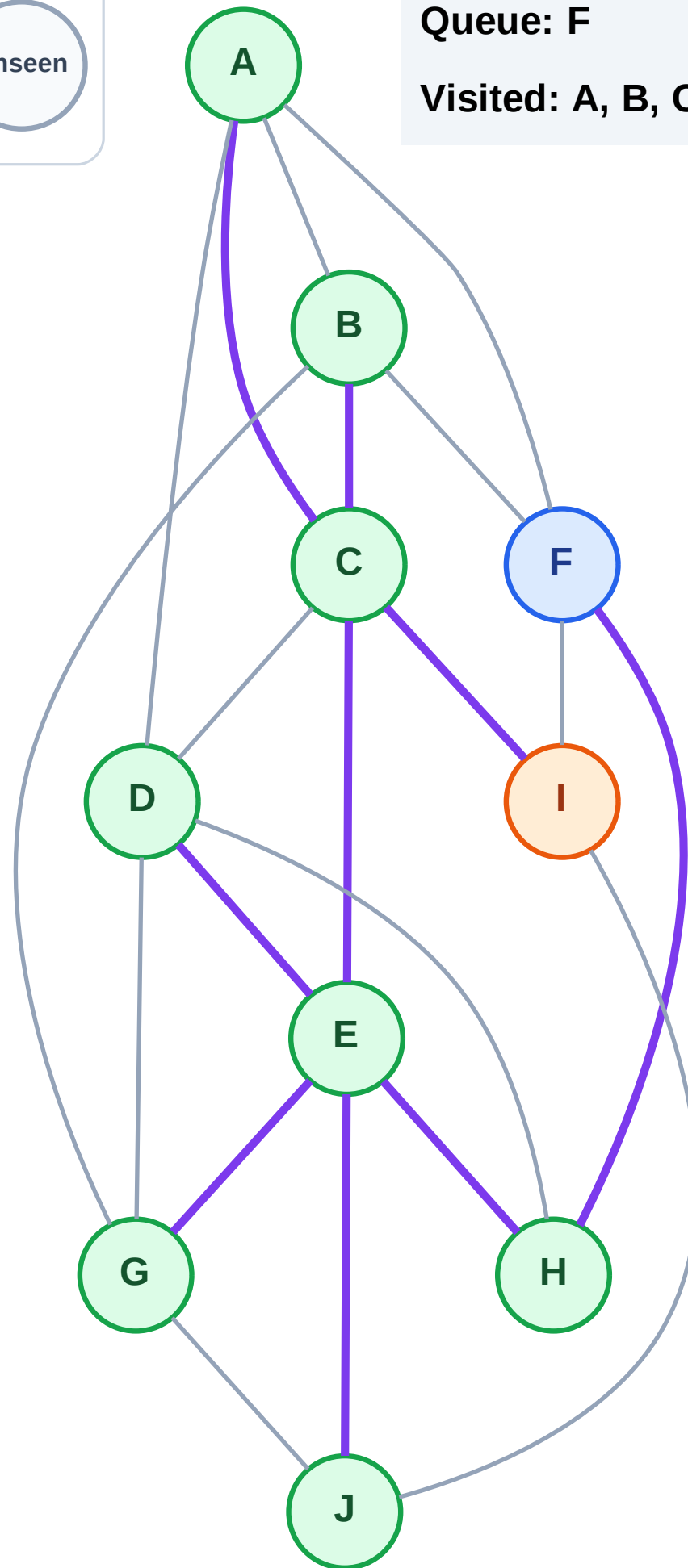
Step 18: Process node I

## Legend



Queue: F

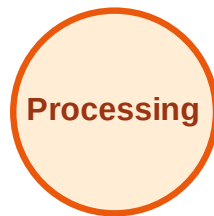
Visited: A, B, C, D, E, G, H, J



# BFS Traversal

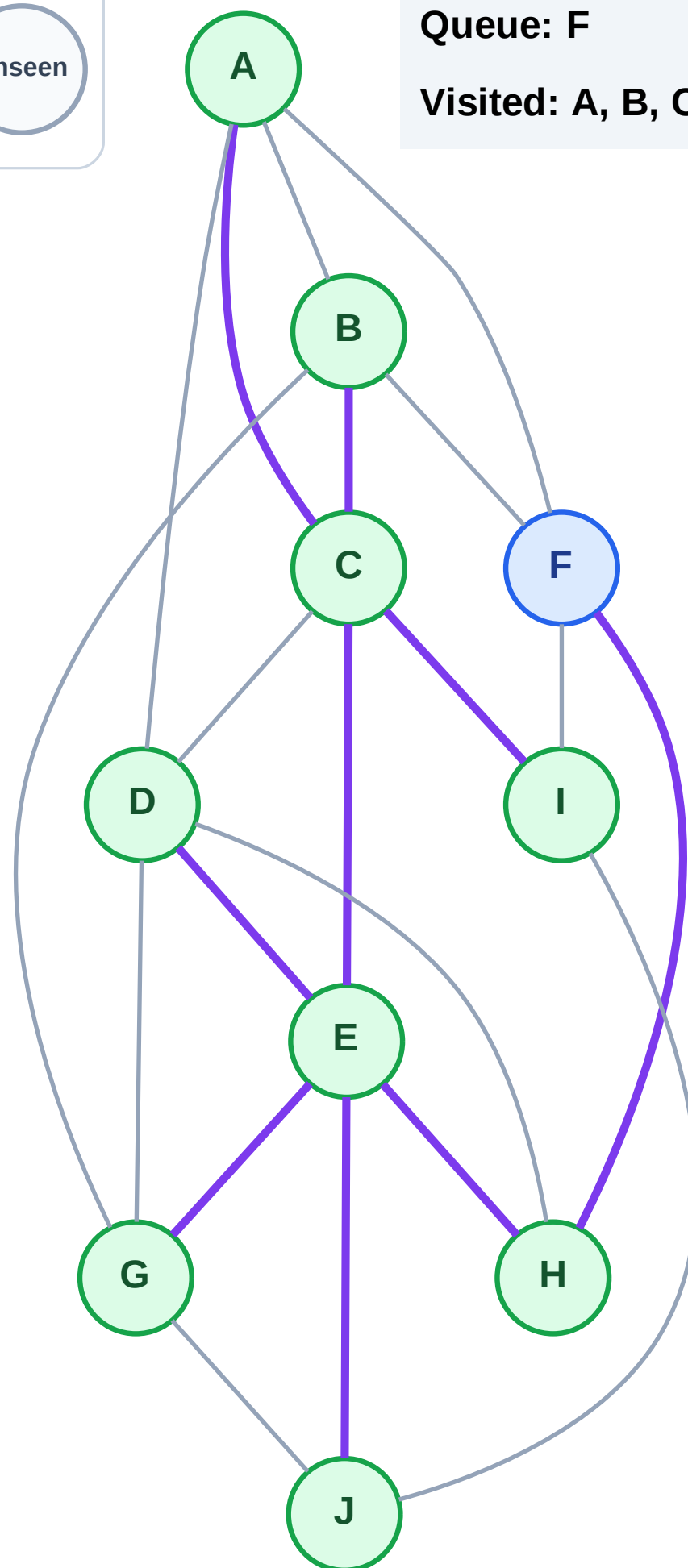
Step 19: No new neighbors from I

## Legend



Queue: F

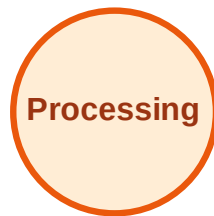
Visited: A, B, C, D, E, G, H, I, J



# BFS Traversal

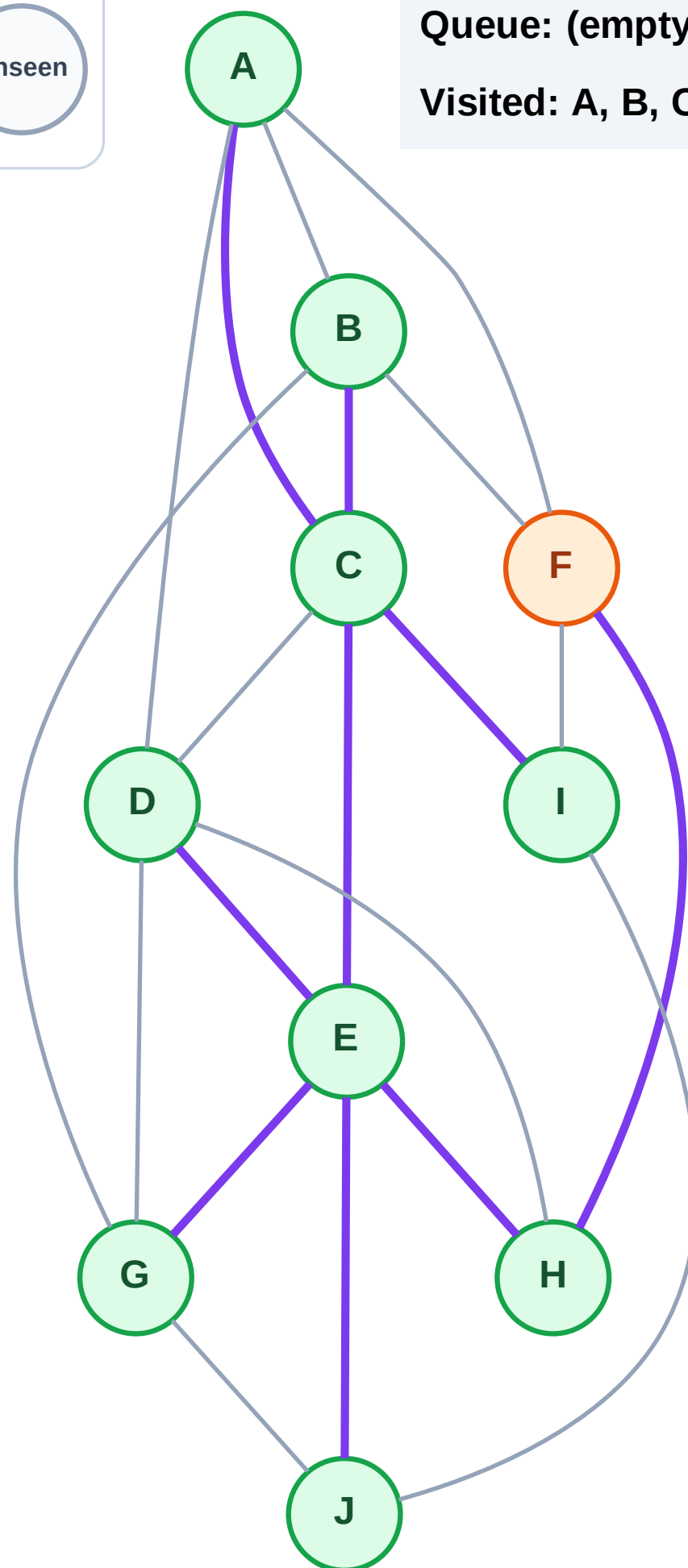
Step 20: Process node F

## Legend



Queue: (empty)

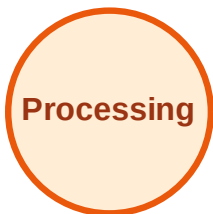
Visited: A, B, C, D, E, G, H, I, J



# BFS Traversal

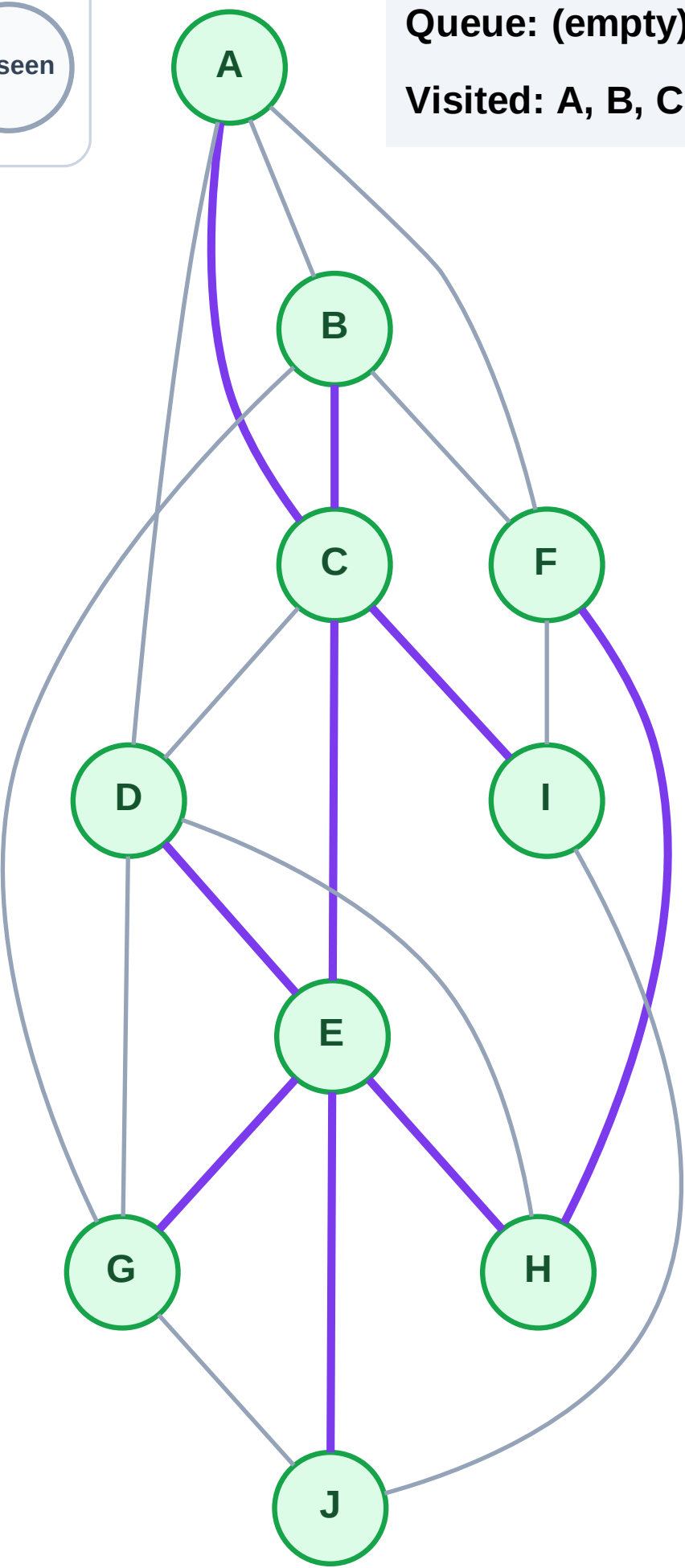
Step 21: No new neighbors from F

## Legend



Queue: (empty)

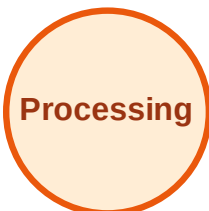
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

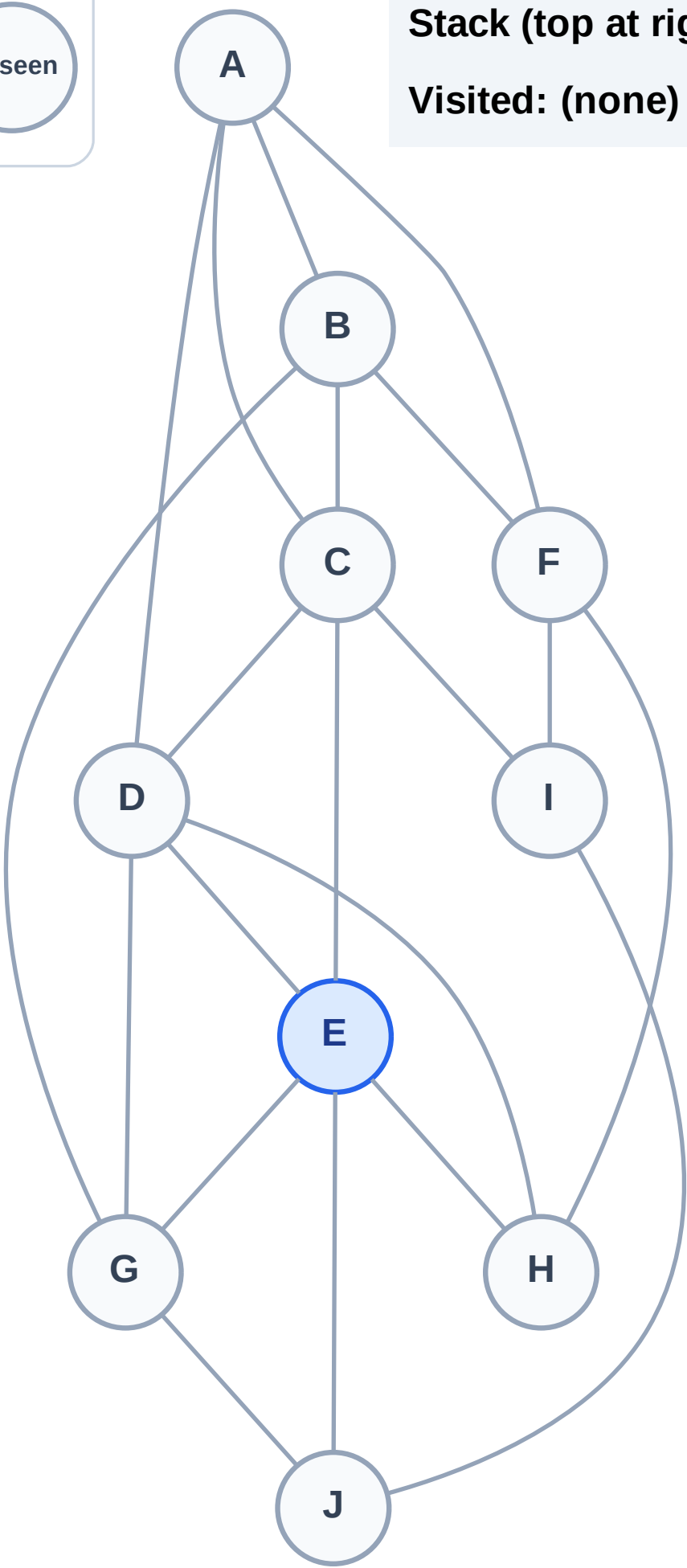
Step 1: Start at E

## Legend



Stack (top at right): E

Visited: (none)





# DFS Traversal

Step 2: Process node E

Legend

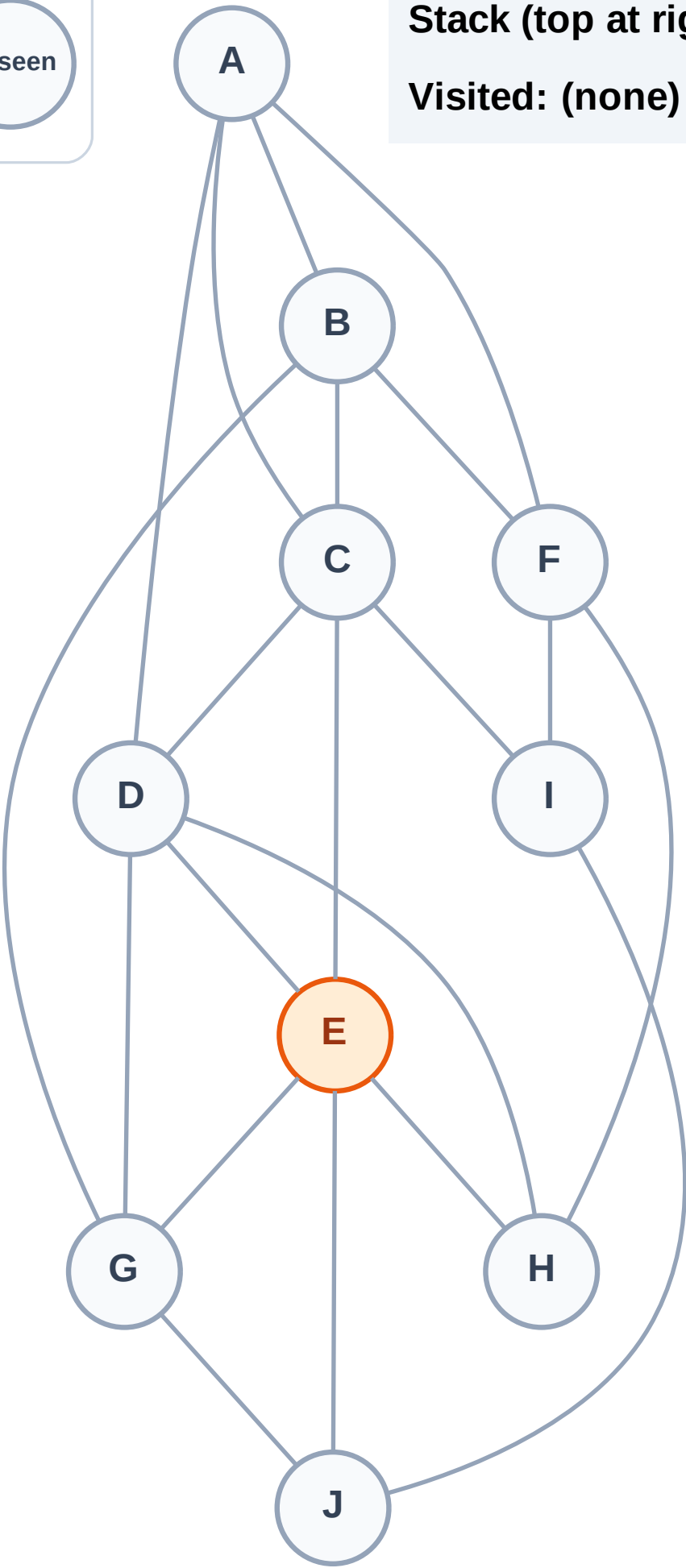
Complete

Processing

Discovered

Unseen

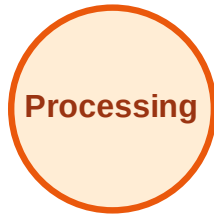
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

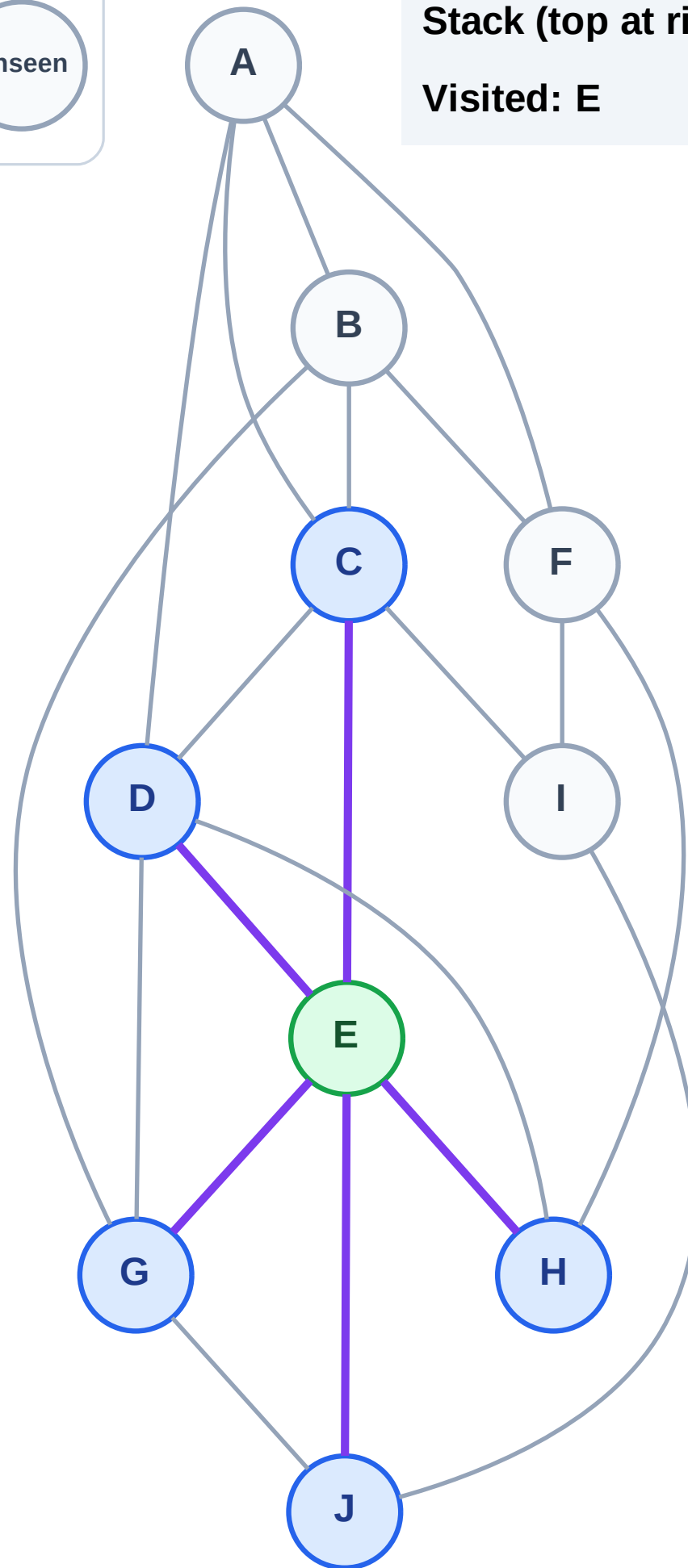
Step 3: Discover J, H, G, D, C from E

## Legend



Stack (top at right): J | H | G | D | C

Visited: E





# DFS Traversal

Step 4: Process node C

Legend

Complete

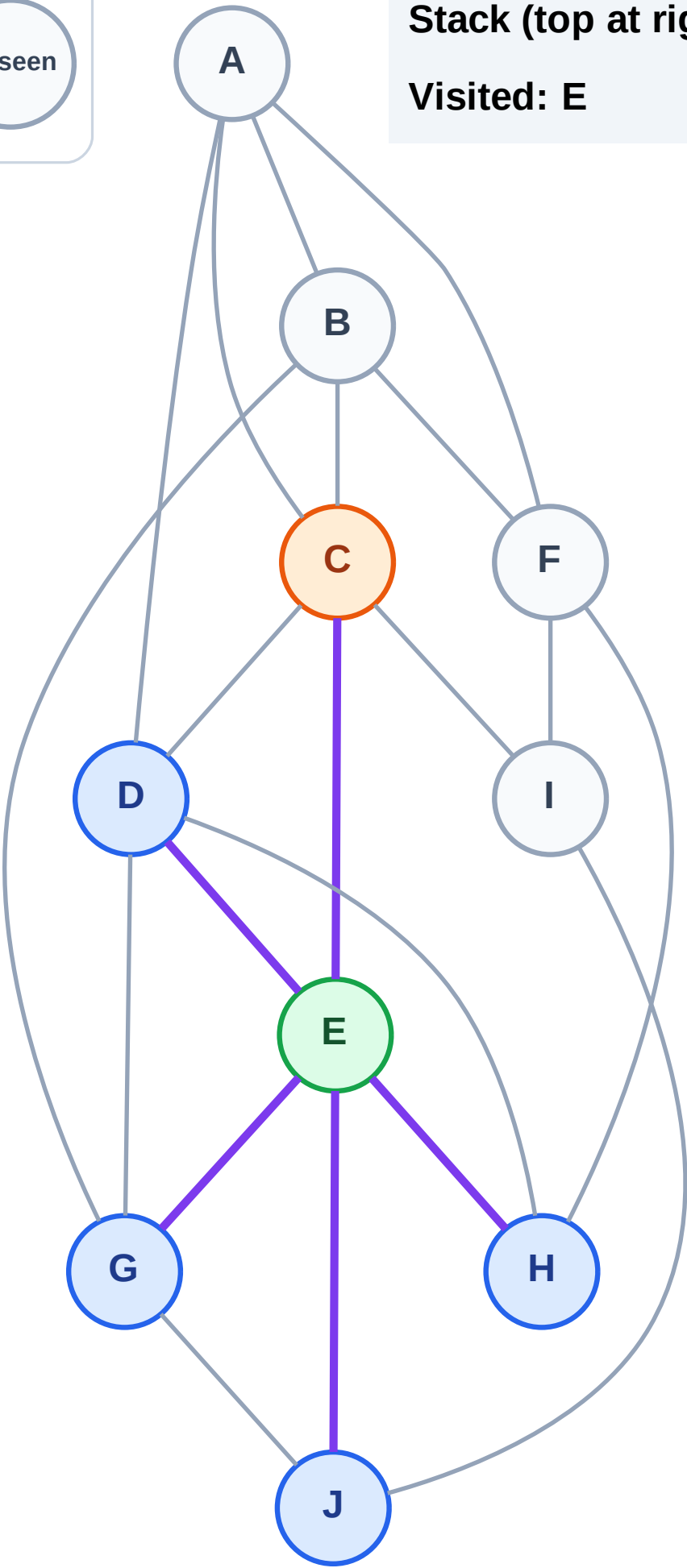
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D

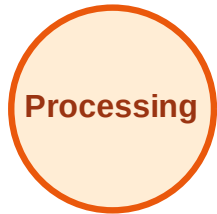
Visited: E



# DFS Traversal

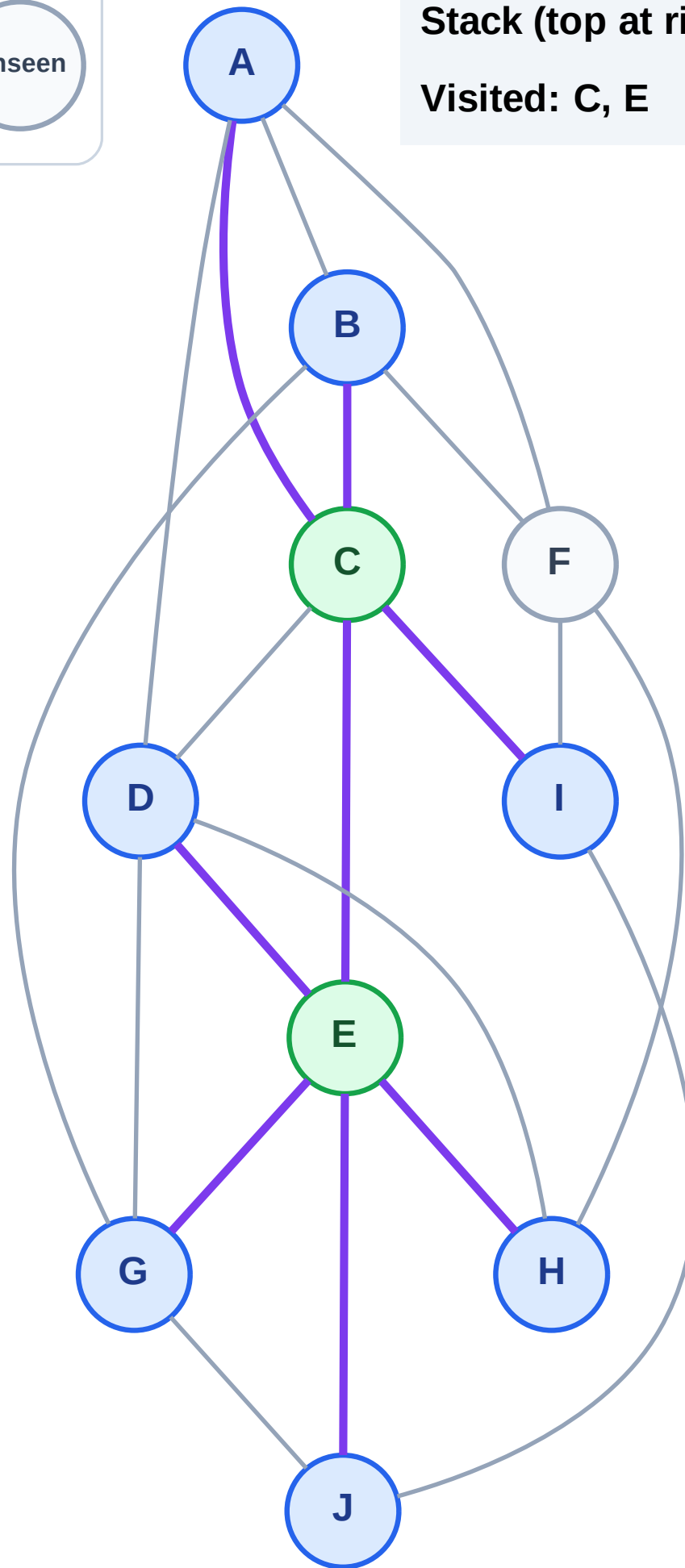
Step 5: Discover I, B, A from C

## Legend



Stack (top at right): J | H | G | D | I | B | A

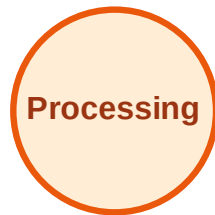
Visited: C, E



# DFS Traversal

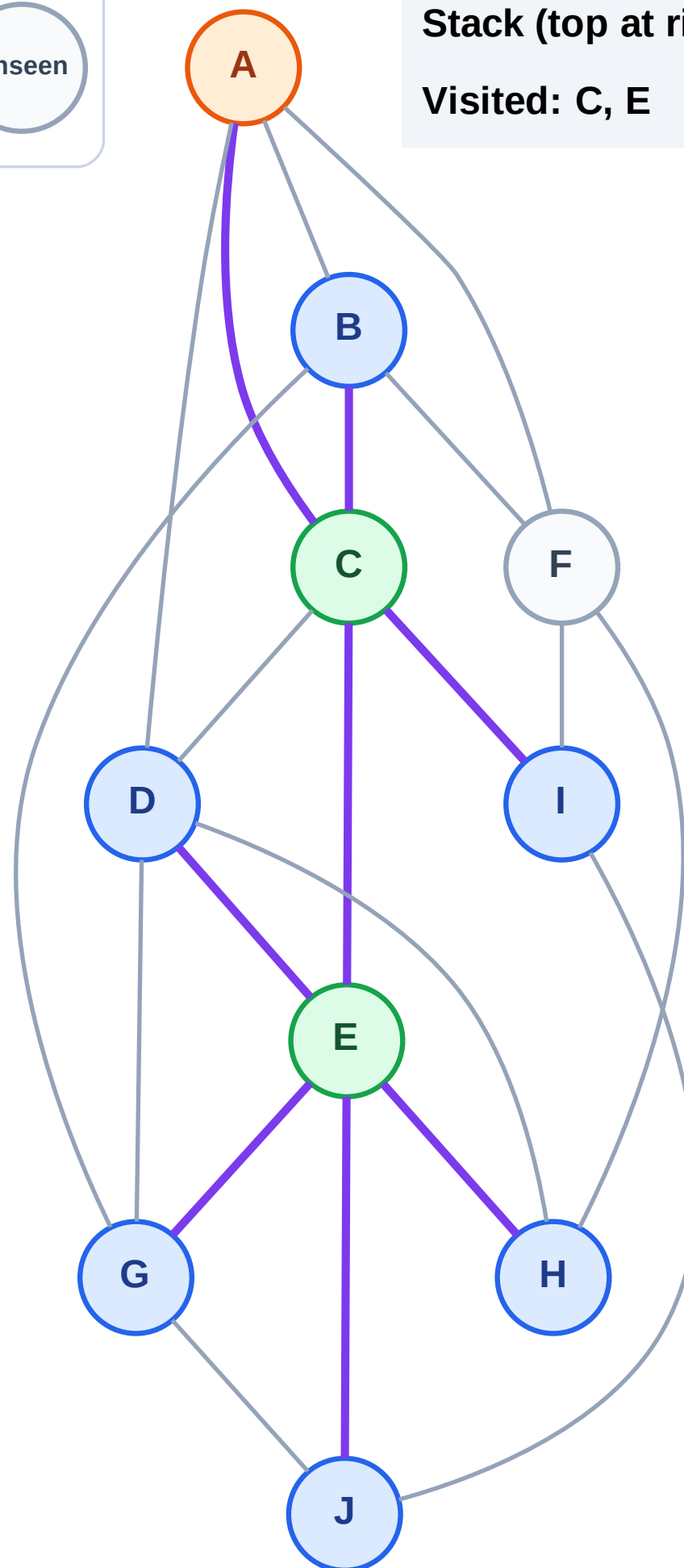
Step 6: Process node A

## Legend



Stack (top at right): J | H | G | D | I | B

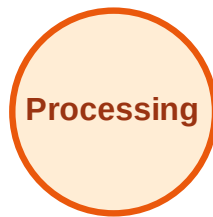
Visited: C, E



# DFS Traversal

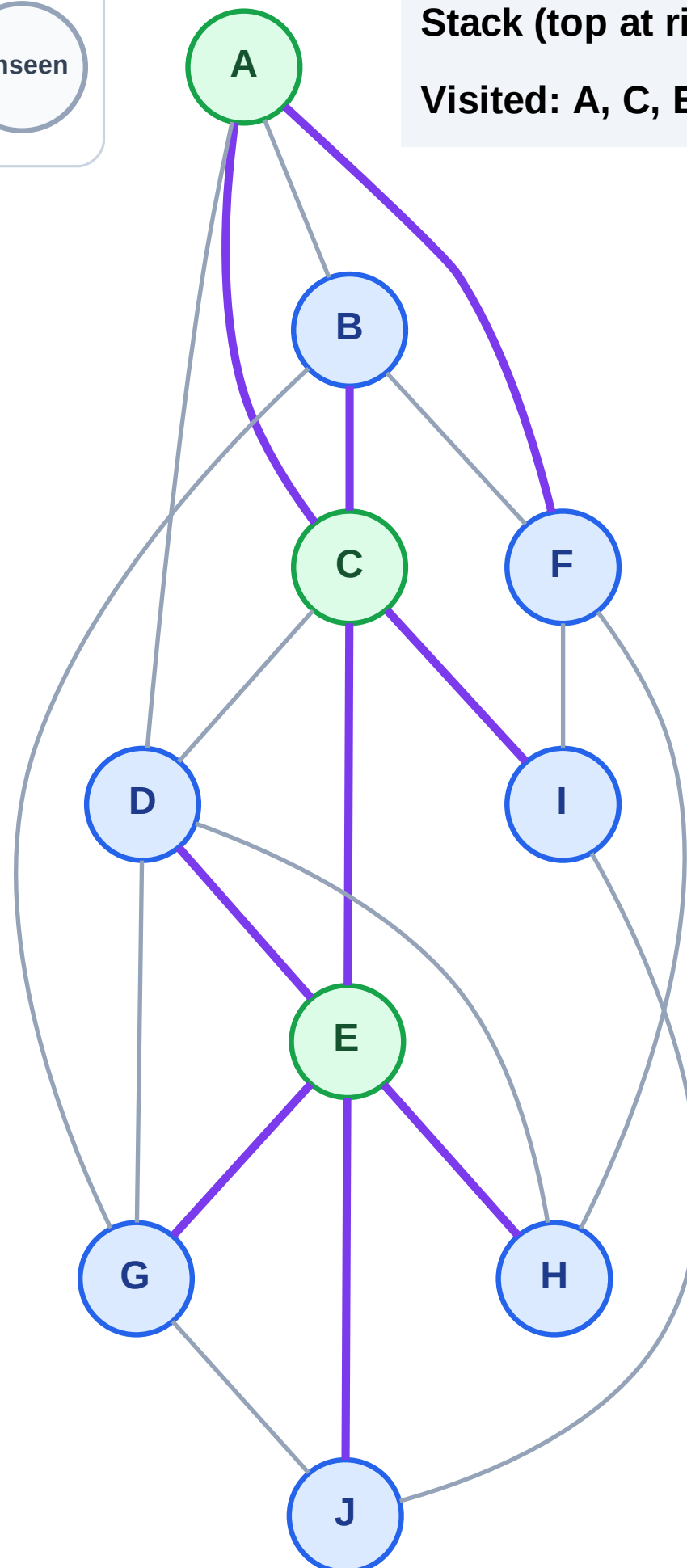
Step 7: Discover F from A

## Legend



Stack (top at right): J | H | G | D | I | B | F

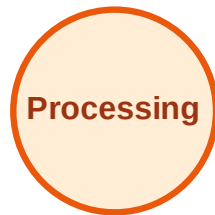
Visited: A, C, E



# DFS Traversal

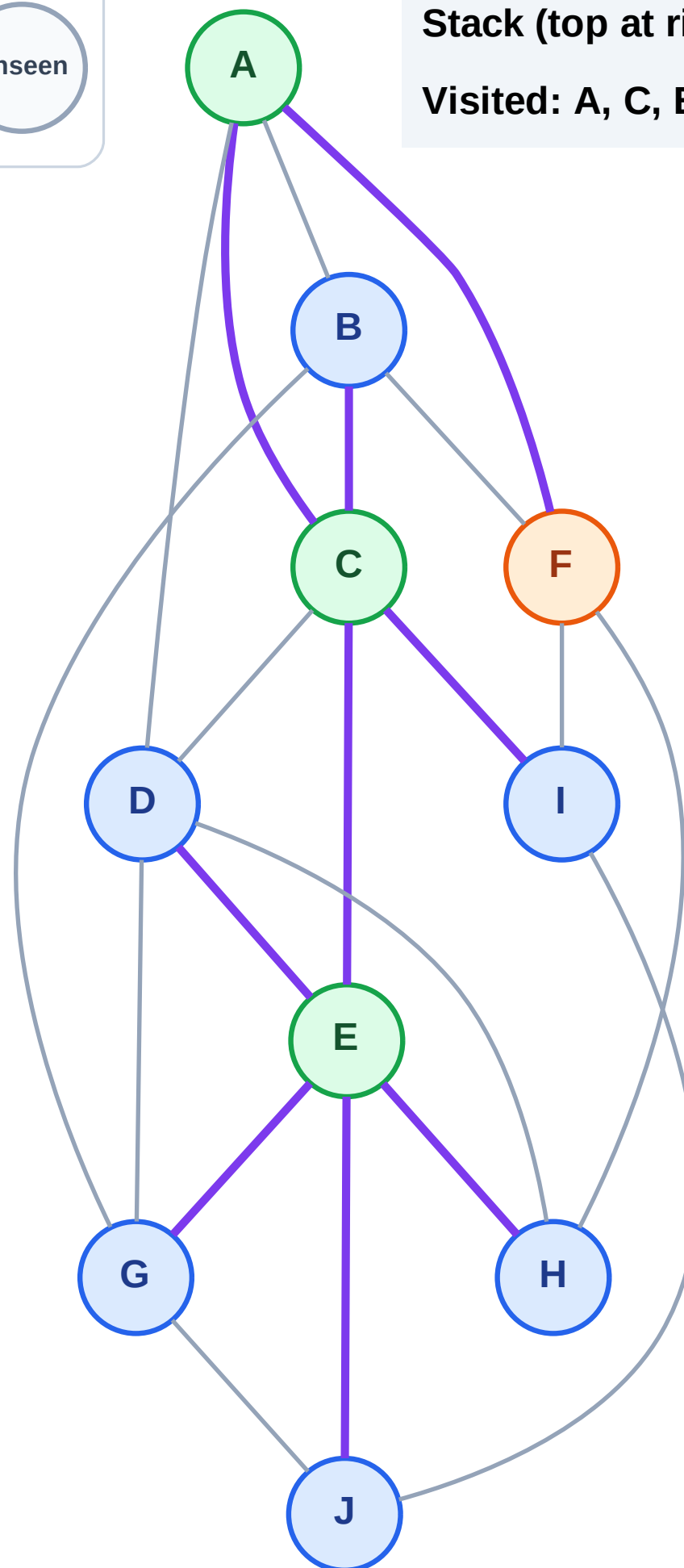
Step 8: Process node F

## Legend



Stack (top at right): J | H | G | D | I | B

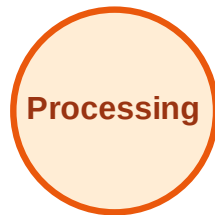
Visited: A, C, E



# DFS Traversal

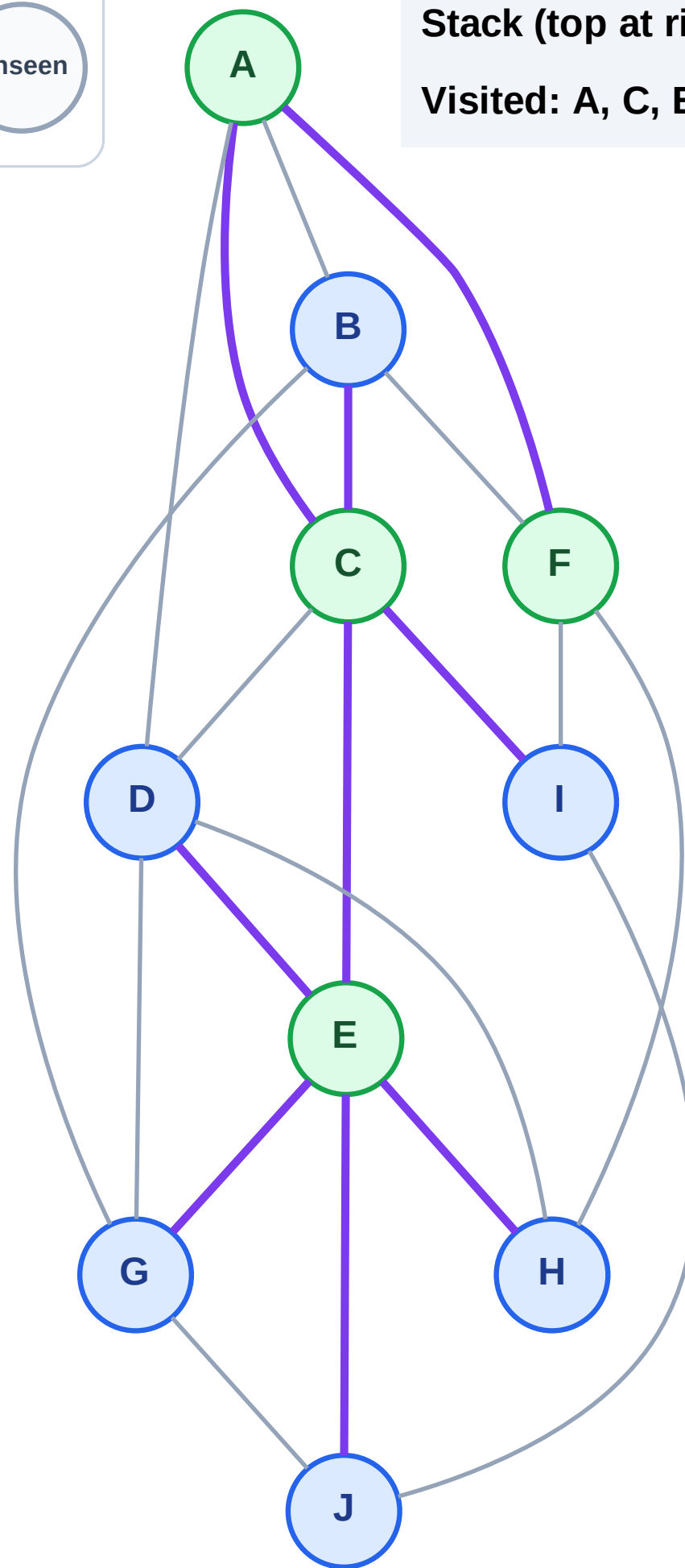
Step 9: No new neighbors from F

## Legend



Stack (top at right): J | H | G | D | I | B

Visited: A, C, E, F





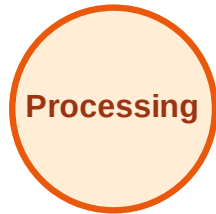
# DFS Traversal

Step 10: Process node B

## Legend



Complete



Processing



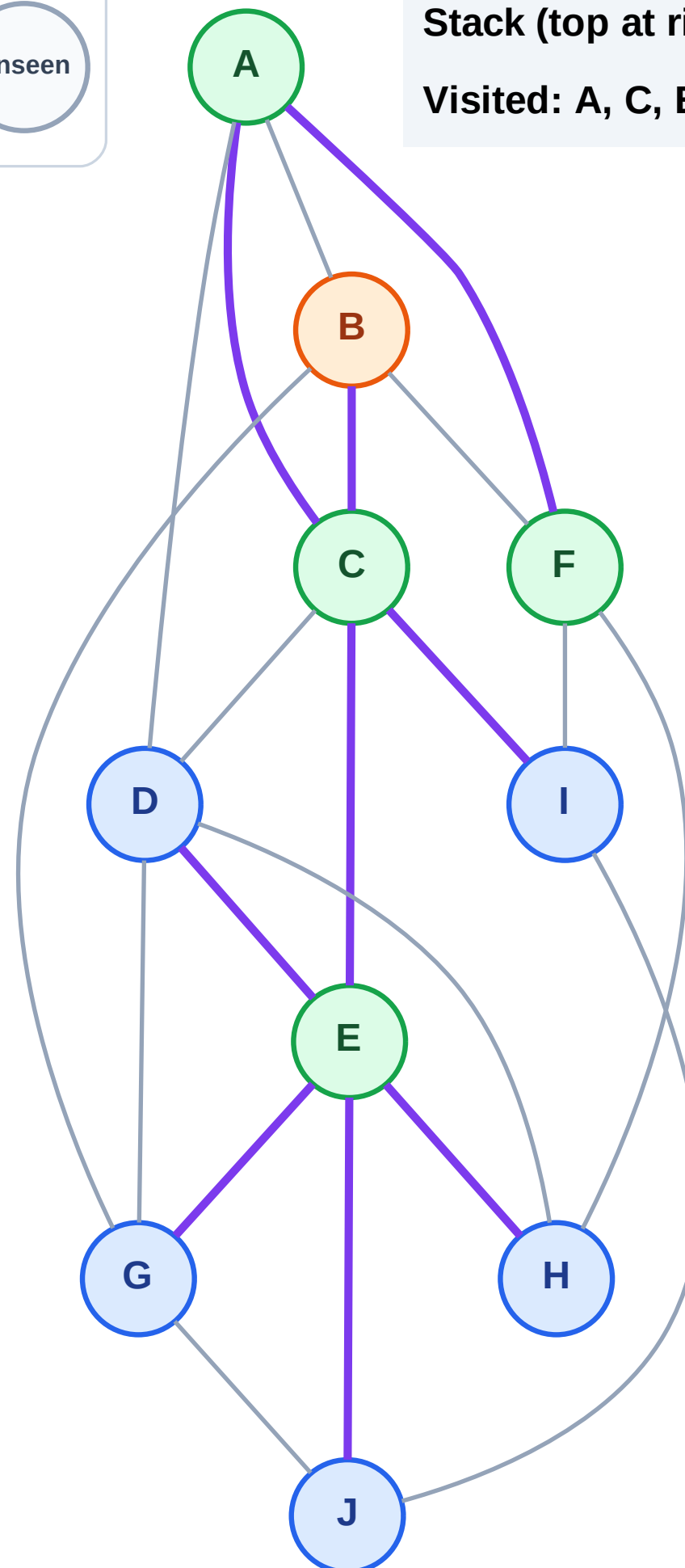
Discovered



Unseen

Stack (top at right): J | H | G | D | I

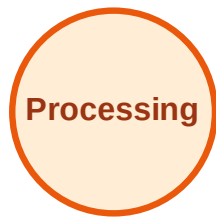
Visited: A, C, E, F



# DFS Traversal

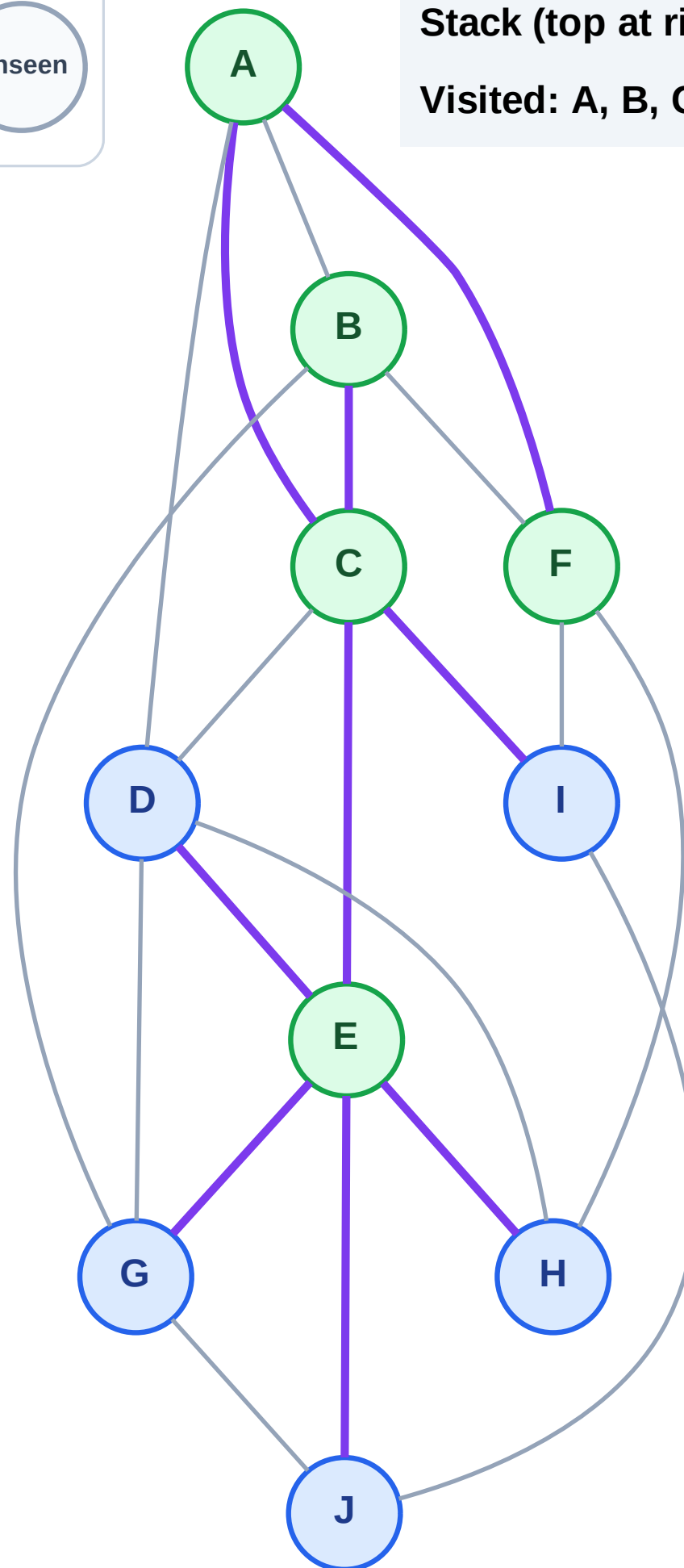
Step 11: No new neighbors from B

## Legend



Stack (top at right): J | H | G | D | I

Visited: A, B, C, E, F

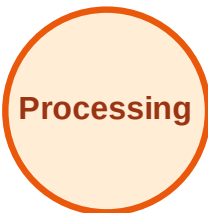




# DFS Traversal

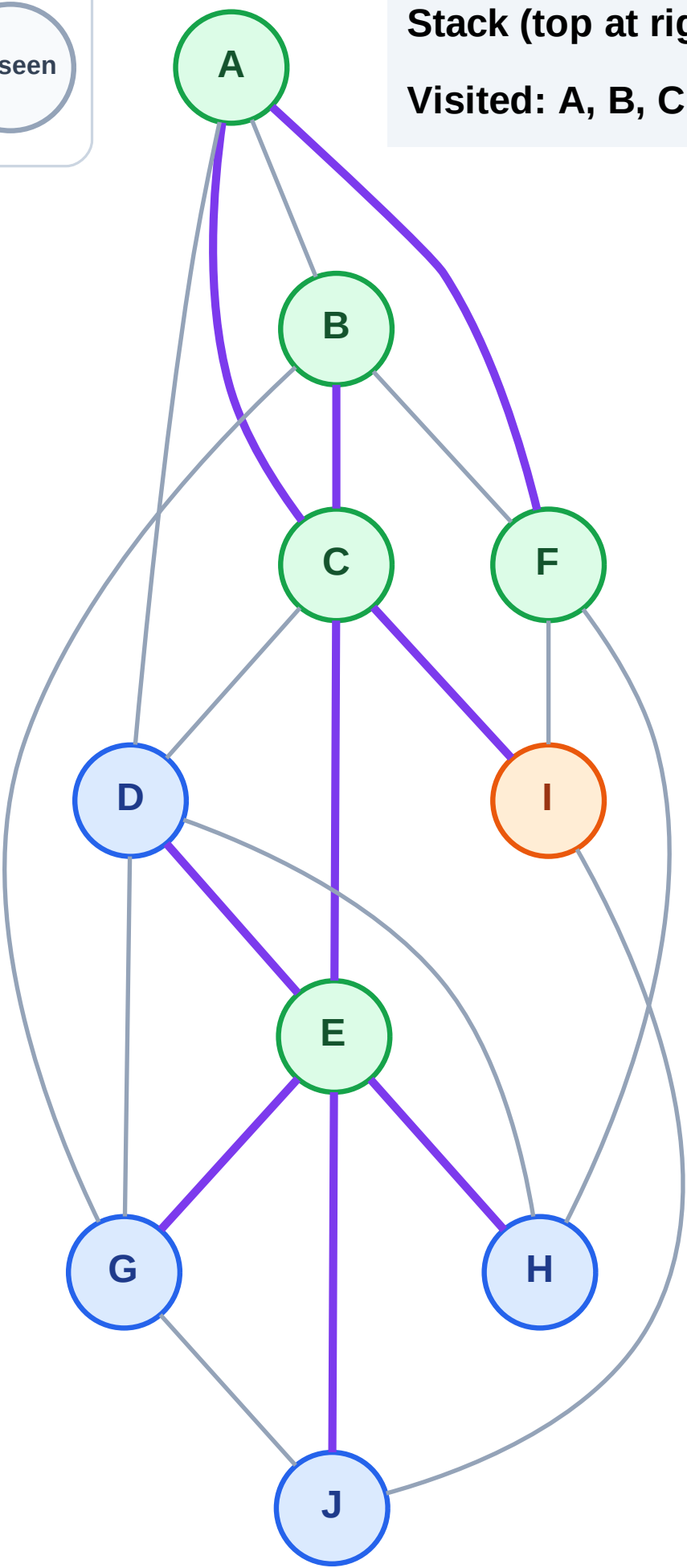
Step 12: Process node I

## Legend



Stack (top at right): J | H | G | D

Visited: A, B, C, E, F



# DFS Traversal

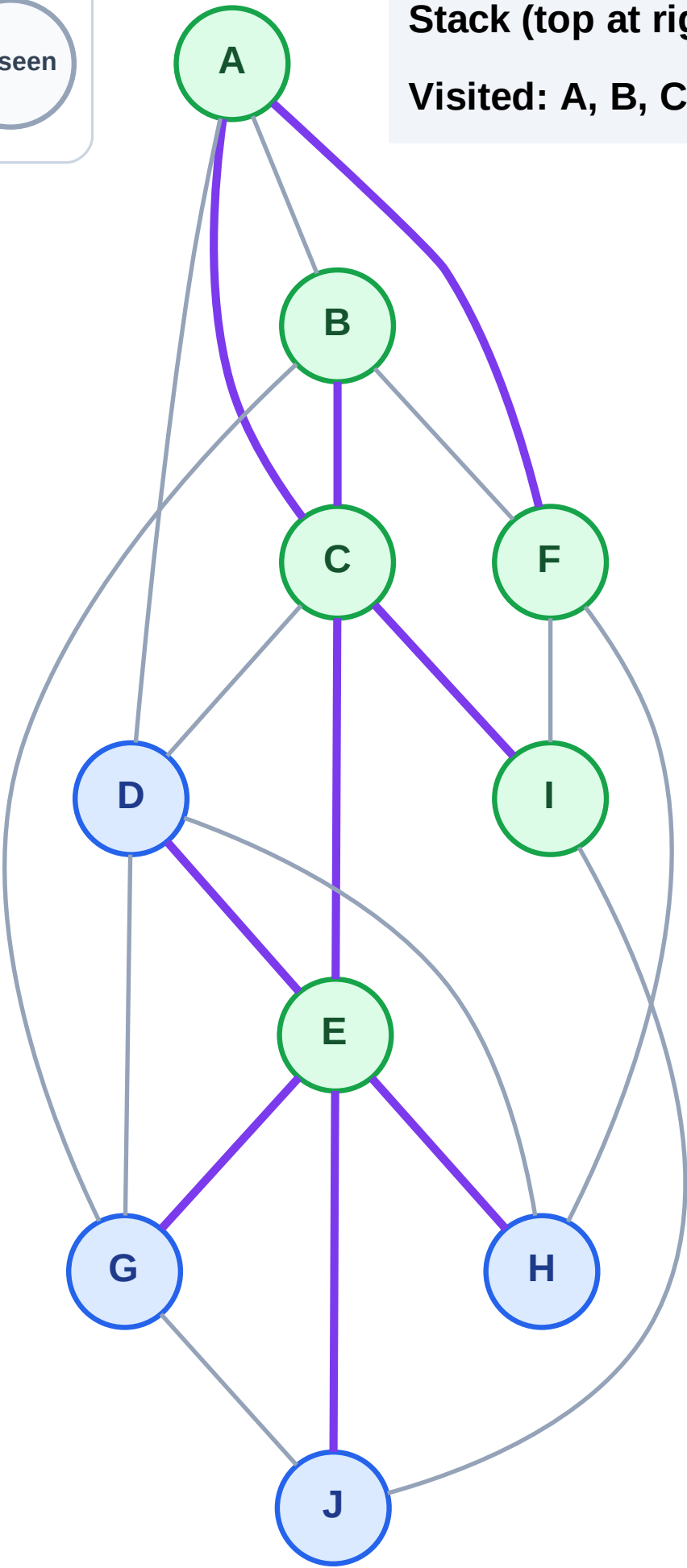
Step 13: No new neighbors from I

## Legend



Stack (top at right): J | H | G | D

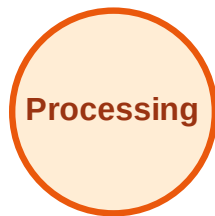
Visited: A, B, C, E, F, I



# DFS Traversal

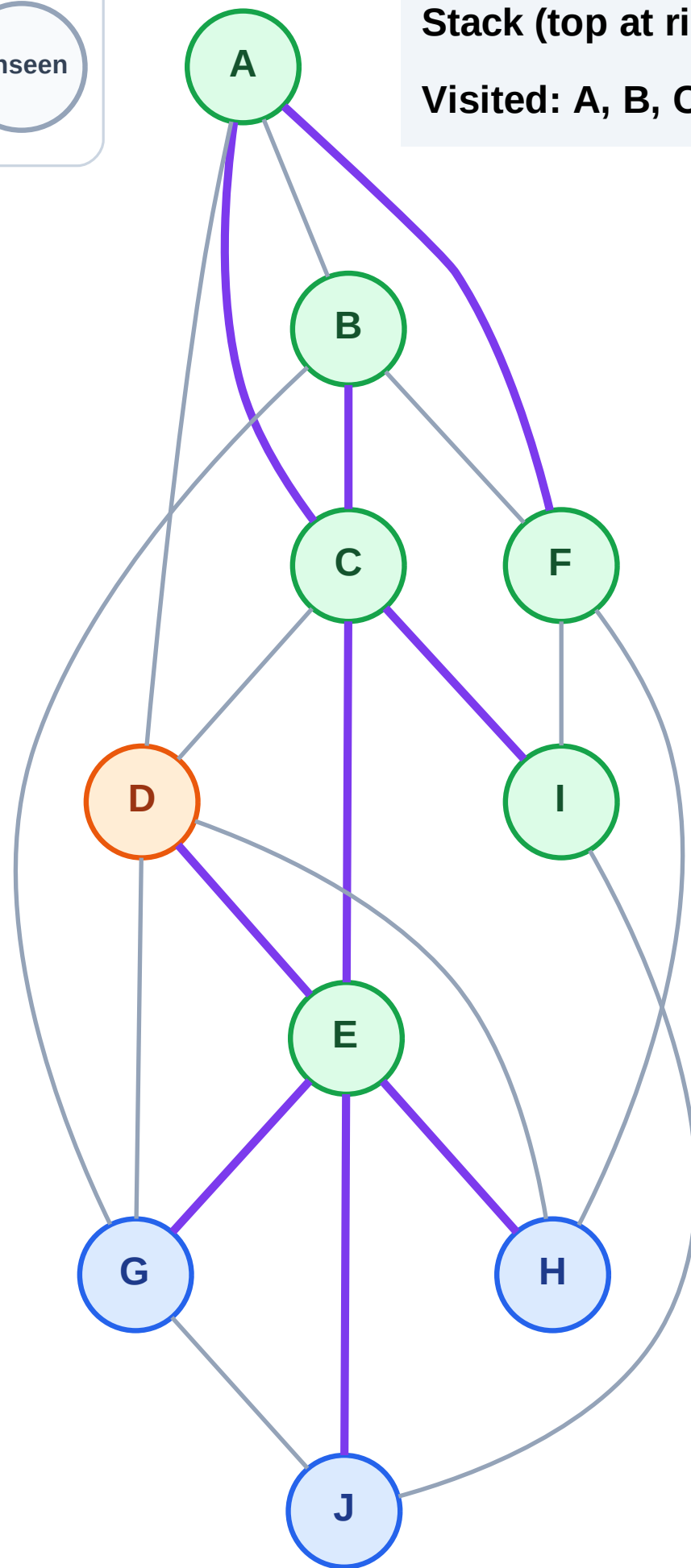
Step 14: Process node D

## Legend



Stack (top at right): J | H | G

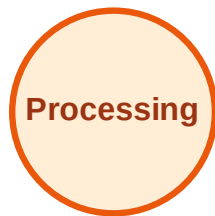
Visited: A, B, C, E, F, I



# DFS Traversal

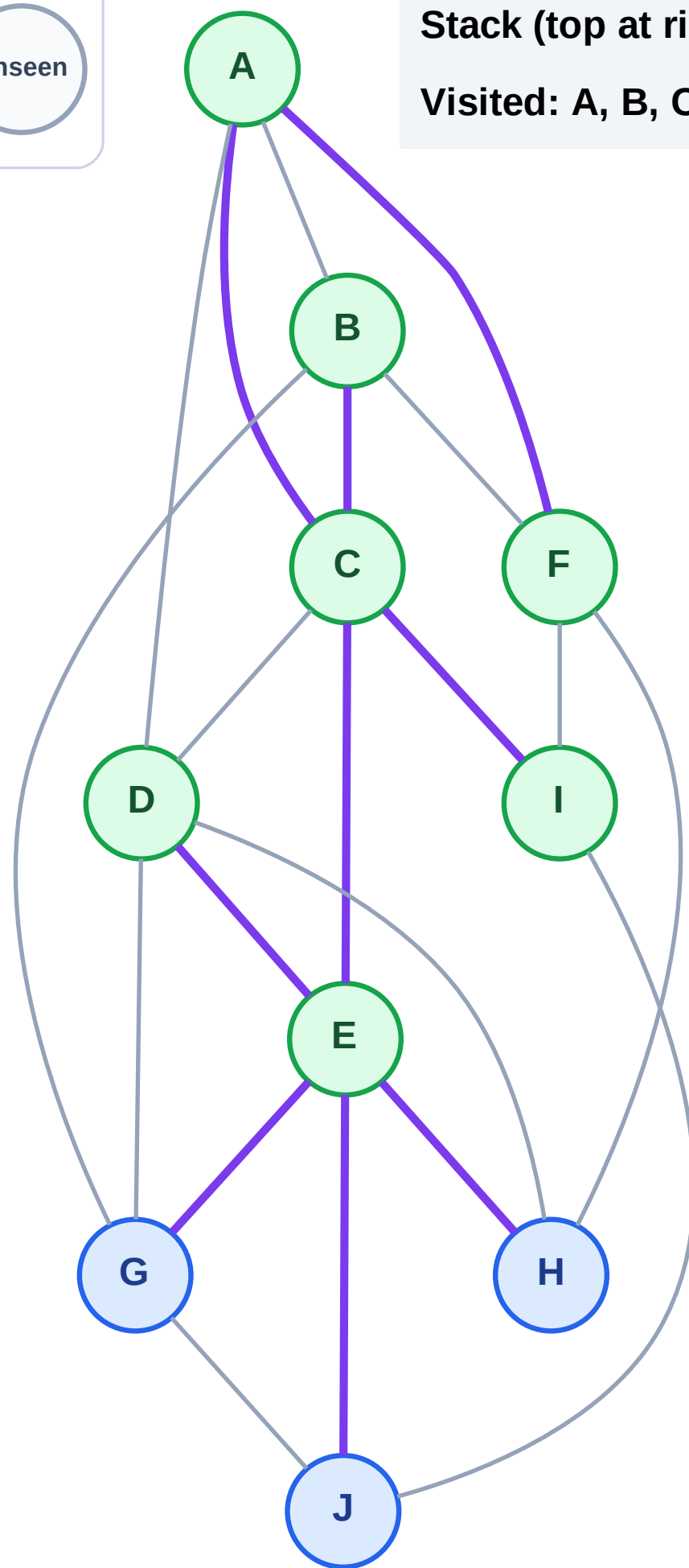
Step 15: No new neighbors from D

## Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



# DFS Traversal

Step 16: Process node G

## Legend

Complete

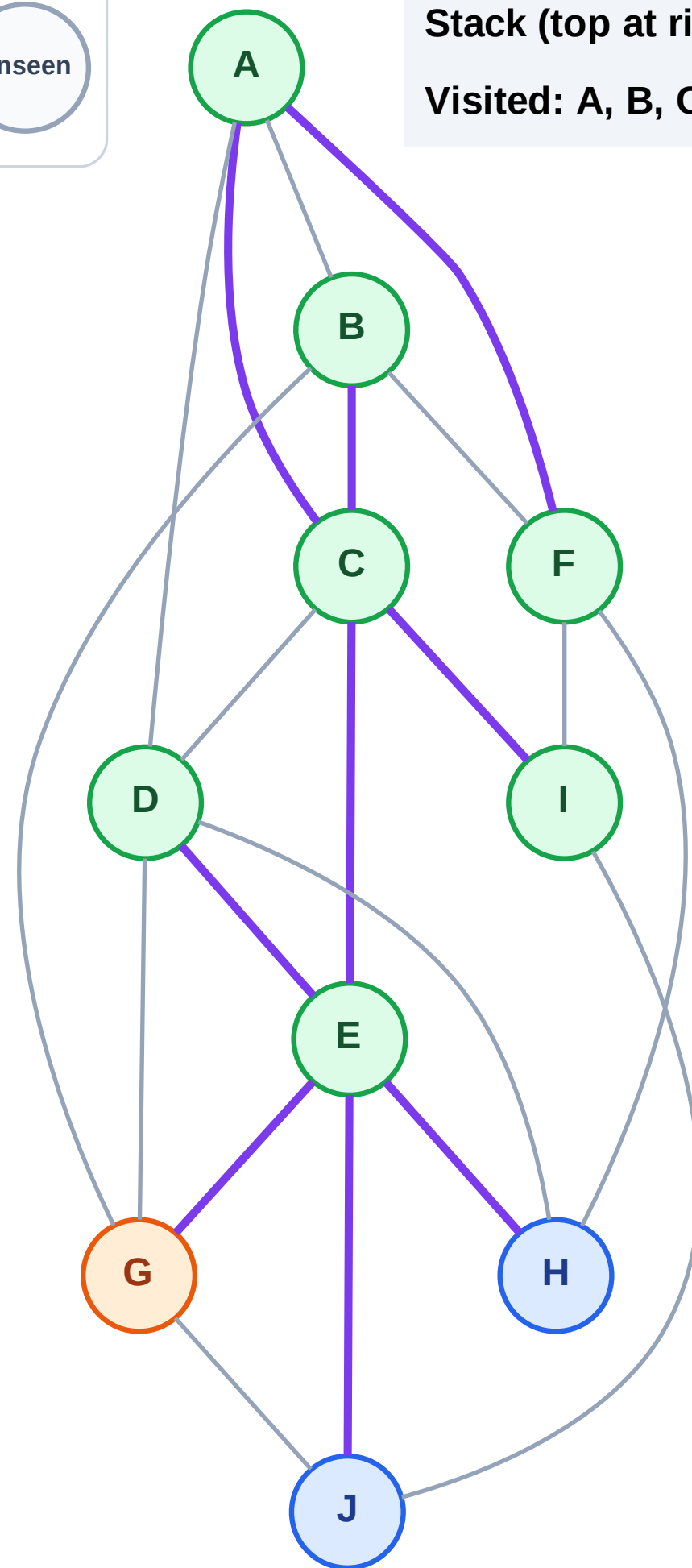
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



# DFS Traversal

Step 17: No new neighbors from G

## Legend

Complete

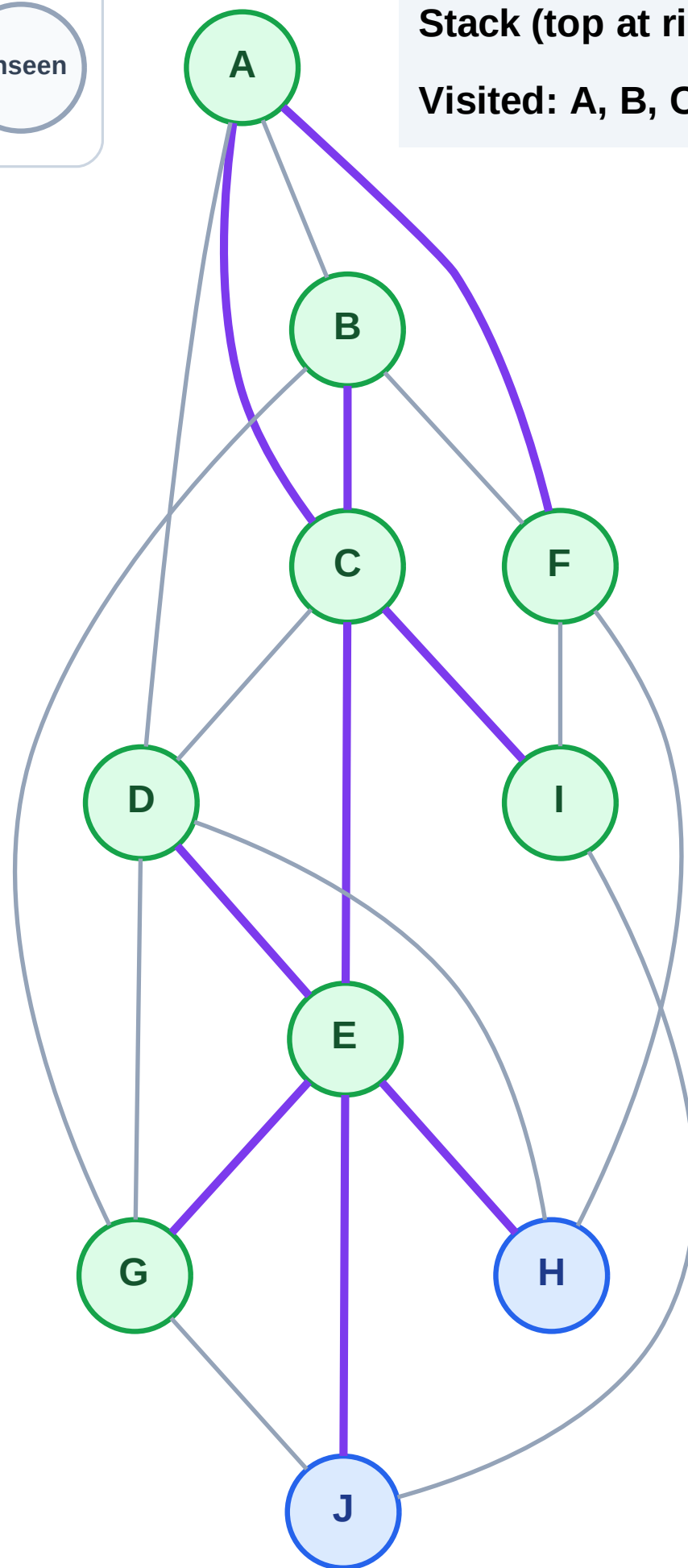
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I





# DFS Traversal

Step 18: Process node H

Legend

Complete

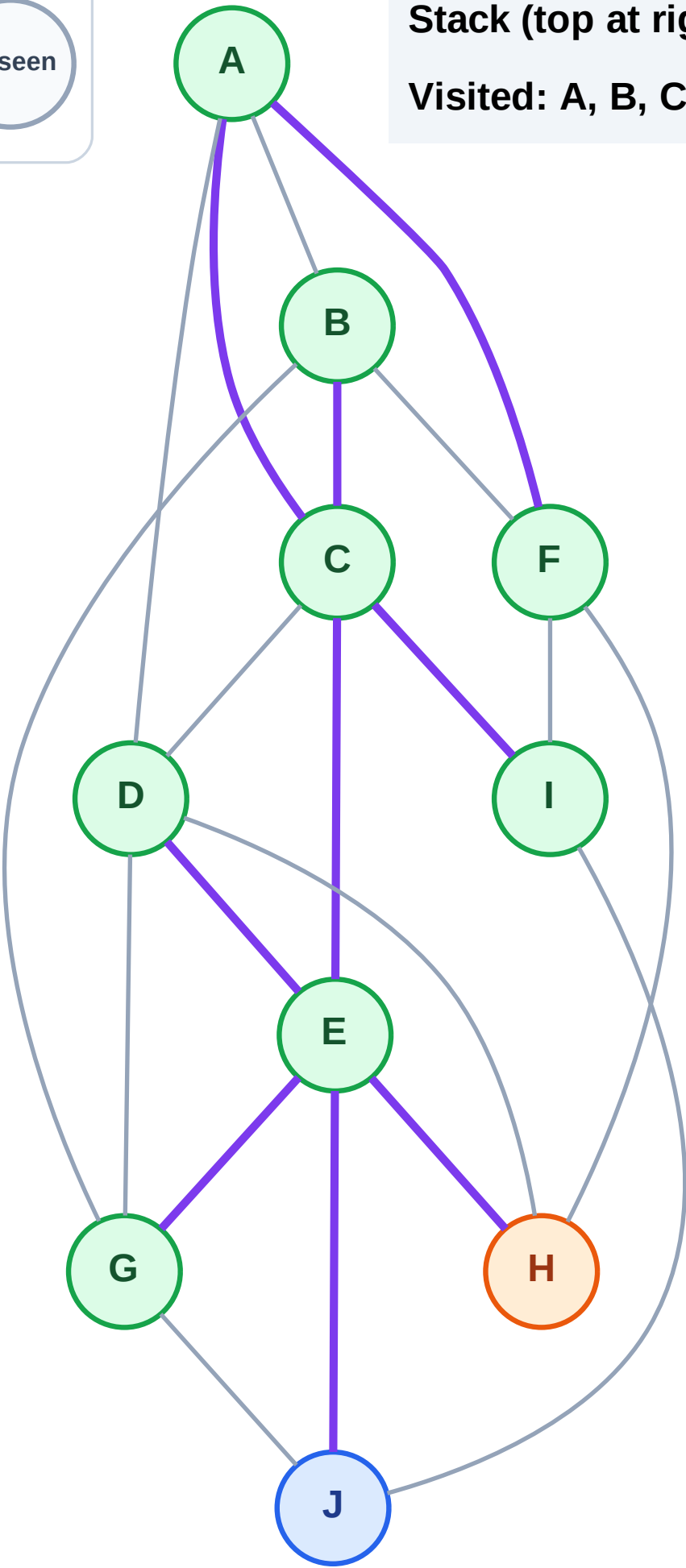
Processing

Discovered

Unseen

Stack (top at right): J

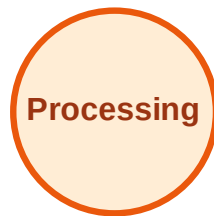
Visited: A, B, C, D, E, F, G, I



# DFS Traversal

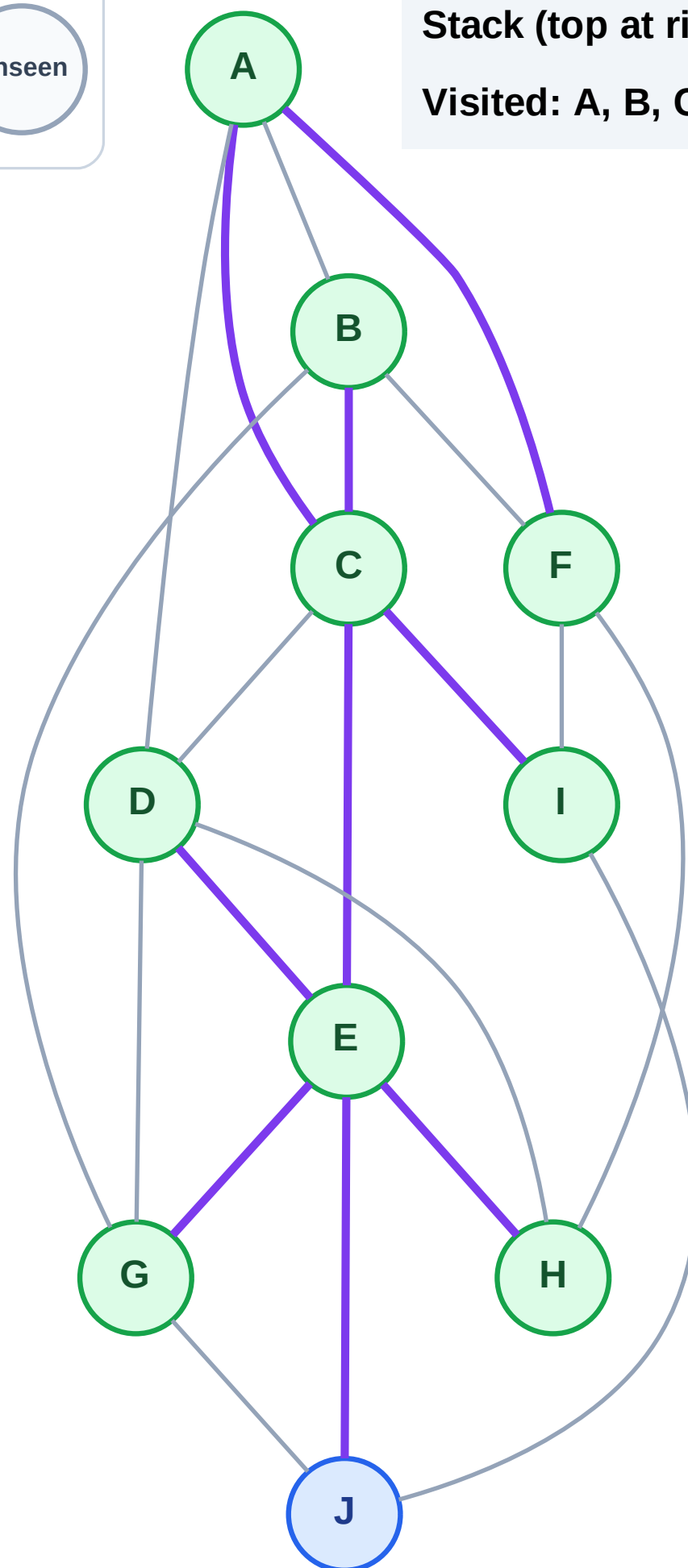
Step 19: No new neighbors from H

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

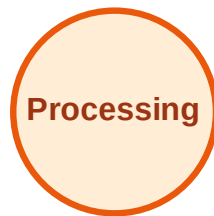




# DFS Traversal

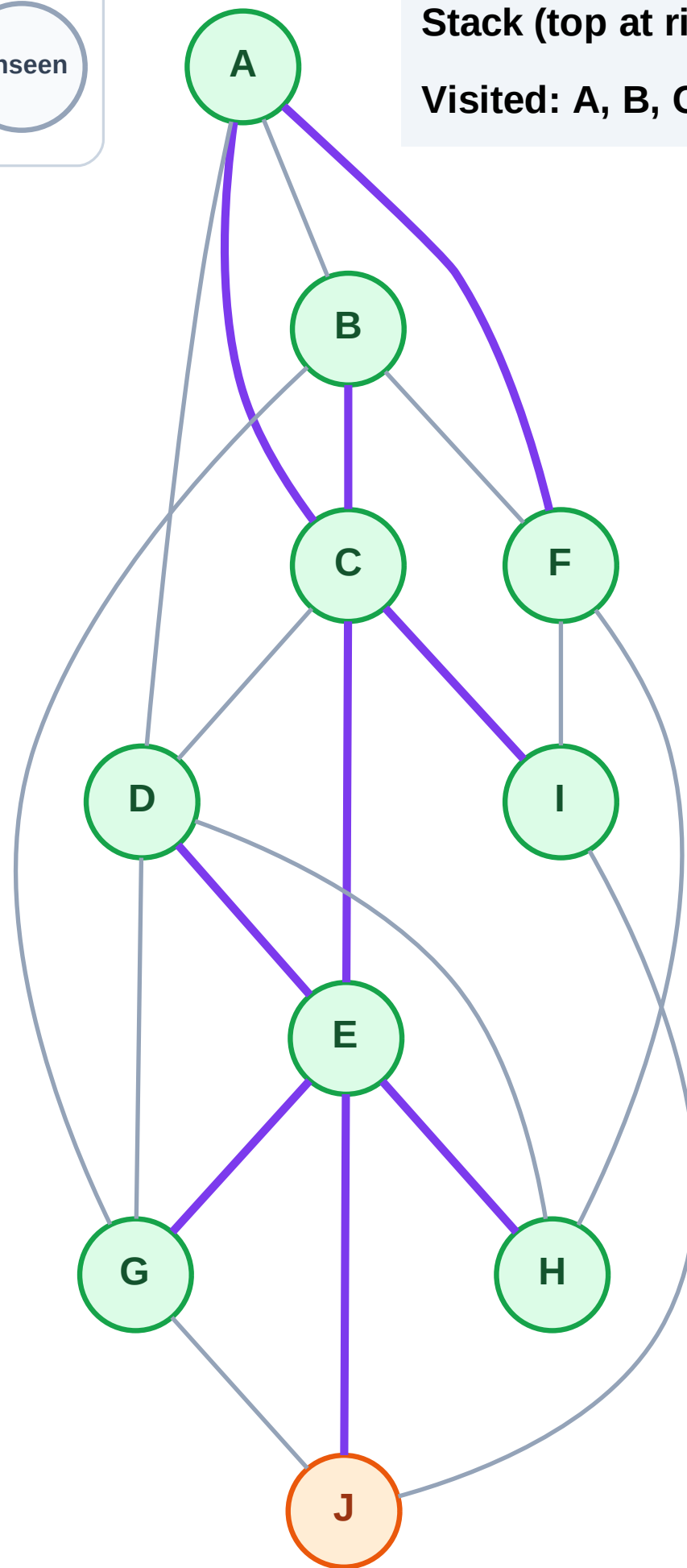
Step 20: Process node J

## Legend



Stack (top at right): (empty)

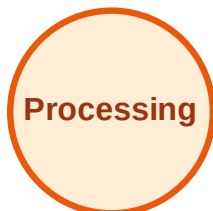
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

